

# Pages 1 - 10: Probability Fundamentals and Axioms

- **1. Basic Concepts of Probability**

- **Random Experiment:** Defined by three conditions:
  - a. All possible outcomes are known in advance.
  - b. The specific outcome of a single trial is not known in advance.
  - c. The experiment can be repeated under identical conditions.
- **Sample Space ( $S$ ):** The set of all possible outcomes of a random experiment.
- **Sample Point:** A single point within the sample space.
- **Event:** A subset of the sample space. If the outcome of an experiment belongs to an event  $A$ , event  $A$  has occurred.
- **Mutually Exclusive Events:** Two events  $A$  and  $B$  that cannot occur together, meaning their intersection is empty:  $A \cap B = \phi$ .
- **Exhaustive Events:** A set of events  $A_1, A_2, \dots$  whose union covers the entire sample space:  $\cup A_i = S$ .

- **2. Definitions of Probability**

- **Classical Definition:** If an experiment yields  $n$  mutually exclusive and equally likely outcomes, and  $n_A$  of these are in event  $A$ , the probability is  $P(A) = \frac{n_A}{n}$  (favorable cases over total cases).
- **Relative Frequency (Statistical) Definition:** If an experiment is repeated  $n$  times and event  $A$  occurs  $n_A$  times, the relative frequency is  $f_A = \frac{n_A}{n}$ . The probability is the limit as  $n$  approaches infinity:  $P(A) = \lim_{n \rightarrow \infty} \frac{n_A}{n}$ .
- **Axiomatic Definition (Kolmogorov, 1933):** Based on set theory and measure theory.
  - **$\sigma$ -field ( $\sigma$ -algebra):** A class  $B$  of subsets of  $S$  satisfying:
    - a.  $\phi \in B$ .
    - b. If  $A \in B$ , then its complement  $A^c \in B$ .
    - c. If  $A_1, A_2, \dots$  is a sequence of sets in  $B$ , then their countable union is also in  $B$ :  
 $\cup_{i=1}^{\infty} A_i \in B$ .
  - **Probability Measure:** A real-valued set function  $P(\cdot)$  on domain  $B$  satisfying three axioms:
    - a.  $P(A) \geq 0$  for all  $A \in B$ .
    - b.  $P(S) = 1$ .
    - c. For a sequence of pairwise disjoint events  $A_1, A_2, \dots$ , the probability of their union is the sum of their probabilities:  $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ .

- **3. Basic Properties of Probability**

- **Impossible Event:**  $P(\phi) = 0$ . (Derived from  $S \cup \phi = S$  where  $S$  and  $\phi$  are disjoint).

- **Complement Rule:**  $P(A^c) = 1 - P(A)$ . (Derived from  $A \cup A^c = S$ ).
- **Subset Rule:** If  $A \subseteq B$ , then  $P(A) \leq P(B)$ .
- **Difference Rule:**  $P(A \cap B^c) = P(A) - P(A \cap B)$ .
- **4. Addition Rules**
  - **Two Events:**  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ . This represents the probability that at least one of the events occurs.
  - **General Addition Rule (Inclusion-Exclusion Principle):** For  $n$  events  $A_1, A_2, \dots, A_n$ , the probability of their union expands as:  

$$P(\cup A_i) = \sum P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) \cdots + (-1)^{n-1} P(A_1 \cap A_2 \cdots \cap A_n).$$
- **5. Probability Inequalities**
  - **Boole's Inequality:**  $P(A \cup B) \leq P(A) + P(B)$ . Generally:  $P(\cup A_i) \leq \sum P(A_i)$ .
  - **Bonferroni's Inequality:**  $P(A \cap B) \geq P(A) + P(B) - 1$ .
- **6. Monotone Sequences of Events**
  - **Non-Decreasing Sequence:**  $A_n \subseteq A_{n+1}$  for all  $n$ . The limit is defined as the union:  

$$\lim_{n \rightarrow \infty} A_n = \cup_{n=1}^{\infty} A_n.$$
  - **Non-Increasing Sequence:**  $A_{n+1} \subseteq A_n$  for all  $n$ . The limit is defined as the intersection:  

$$\lim_{n \rightarrow \infty} A_n = \cap_{n=1}^{\infty} A_n.$$
  - **Limit Theorem:** The probability of the limit of a monotone sequence is the limit of the probabilities.
    - For a non-decreasing sequence:  $\lim_{n \rightarrow \infty} P(A_n) = P(\cup_{n=1}^{\infty} A_n)$ .
    - (Proof relies on constructing a new sequence of pairwise disjoint sets  $B_n = A_n - A_{n-1}$ ).
- **7. Conditional Probability (Introduction)**
  - **Concept:** When determining the probability of event  $A$  given that event  $B$  has occurred,  $B$  effectively becomes the "new sample space."
  - **Visualization:** Analyzed by looking at the intersection  $A \cap B$  strictly within the boundaries of event  $B$ .

## Pages 11 - 20: Conditional Probability, Independence, and Random Variables

- **1. Conditional Probability**
  - **Definition:** For a sample space  $S$  and an event  $B$  where  $P(B) > 0$ , the conditional probability of an event  $A$  given  $B$  is defined as  $P(A|B) = \frac{P(A \cap B)}{P(B)}$ . If  $P(B) = 0$ ,  $P(A|B)$  is not defined.

- **Validity as a Probability Measure:** The conditional probability  $P_B(A) = P(A|B)$  satisfies the axioms of a probability measure on a  $\sigma$ -field  $C$ .

- $P_B(A) \geq 0$ .
- $P_B(S) = P(S|B) = \frac{P(S \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$ .
- For pairwise disjoint events  $A_1, A_2, \dots$ , the probability of their union is the sum of their conditional probabilities:  $P_B(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P_B(A_i)$ .

- **2. Multiplication Rule of Probability**

- **Two Events:** If  $P(A), P(B) > 0$ , then  $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$ .
- **Three Events:**  $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)$ .
- **General Case:** For  $n$  events, the intersection probability expands sequentially:  $P(A_1 \cap \dots \cap A_n) = P(A_1)P(A_2|A_1) \dots P(A_n|A_1 \cap \dots \cap A_{n-1})$ .

- **3. Theorem of Total Probability**

- **Setup:** Let  $E_1, E_2, \dots, E_n$  be pairwise disjoint and exhaustive events forming the sample space  $S$  ( $E_1 \cup E_2 \cup \dots \cup E_n = S$ ), with  $P(E_i) > 0$  for all  $i$ .
- **Formula:** For any event  $A$  in the sample space,  $P(A) = \sum_{i=1}^n P(E_i)P(A|E_i)$ .
- **Proof Logic:** Event  $A$  can be expressed as the union of its intersections with the mutually exclusive partitions:  $A = \cup_{i=1}^n (A \cap E_i)$ .

- **4. Bayes's Formula**

- **Formula:** Under the same conditions as Total Probability, and for an event  $A$  where  $P(A) > 0$ , the reverse conditional probability is  $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{i=1}^n P(E_i)P(A|E_i)}$ .
- **Application:** Allows calculating  $P(E_i|A)$  (e.g., probability a defective item came from a specific distributor) by using the prior probabilities  $P(E_i)$  and conditional probabilities  $P(A|E_i)$ .

- **5. Independence of Events**

- **Concept:** Two events  $A$  and  $B$  are independent if the occurrence of one does not affect the occurrence of the other.
- **Condition for Two Events:**  $A$  and  $B$  are independent if  $P(A \cap B) = P(A)P(B)$ . This implies  $P(A|B) = P(A)$  and  $P(B|A) = P(B)$  if probabilities are  $> 0$ .
- **Condition for Three Events:**  $A, B, C$  are independent if they are pairwise independent ( $P(A \cap B) = P(A)P(B)$ , etc.) AND mutually independent:  $P(A \cap B \cap C) = P(A)P(B)P(C)$ .
- **Complements:** If events  $A$  and  $B$  are independent, their complements are also independent relationships (e.g.,  $A^c$  and  $B$  are independent).

- **6. Random Variables**

- **Definition:** A random variable  $X$  is a function mapping the sample space to real numbers ( $X : S \rightarrow \mathbb{R}$ ). For every  $x \in \mathbb{R}$ , the set  $\{w \in S \mid X(w) \leq x\}$  must be an event in the  $\sigma$ -field  $C$ .

- **Induced Measure:** A random variable induces a new probability distribution on  $\mathbb{R}$ , denoted as  $P_X(A) = P[\{w \in S \mid X(w) \in A\}] = P[X^{-1}(A)]$ .
- **Discrete Random Variable:** A random variable is discrete if its range is a finite or countable set.
- **7. Distribution Function (CDF)**
  - **Definition:** The cumulative distribution function (cdf)  $F_X(x)$  is defined as  $P(X \leq x)$  or  $P(-\infty < X \leq x)$ .
  - **Key Properties:**
    - a. It is a non-decreasing function.
    - b. Limits at infinity are absolute:  $F_X(-\infty) = 0$  and  $F_X(\infty) = 1$ .
    - c. It is a right-continuous function.

## Pages 21 - 30: Random Variables & Distributions

- **1. Cumulative Distribution Function (cdf) Properties**
  - **Non-Decreasing:**  $F_X(x_1) \leq F_X(x_2)$  for  $x_1 < x_2$ , which follows from the fact that  $F_X(x_2) - F_X(x_1) = P(x_1 < X \leq x_2) \geq 0$ .
  - **Limits at Infinity:**  $F_X(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$ , proven using a decreasing sequence of sets  $A_n = \{X \leq x_n\}$  where  $\lim_{n \rightarrow \infty} A_n = \emptyset$ . Similarly,  $F_X(\infty) = 1$ .
  - **Right-Continuous:**  $F_X$  is right-continuous, meaning  $\lim_{h \rightarrow 0, h > 0} F_X(x + h) = F_X(x)$ .
  - **Interval Probability:** For any real numbers  $a < b$ ,  $P(a < X \leq b) = F_X(b) - F_X(a)$ .
- **2. Point Probabilities & Left Limits**
  - **Left Limit Definition:**  $F_X(x-) = \lim_{h \rightarrow 0, h > 0} F_X(x - h)$ .
  - **Point Probability Theorem:** For any  $x \in \mathbb{R}$ , the probability at a specific point is  $P(X = x) = F_X(x) - F_X(x-)$ .
  - **Discontinuity:** If  $P(X = x) > 0$ , then  $F_X(x-) \neq F_X(x)$ , which implies  $F_X$  is not left-continuous at  $x$  (forming a step or jump in the graph).
- **3. Discrete Random Variables**
  - **Definition:** A random variable  $X$  is discrete if its range is finite or a countable set.
  - **Probability Mass Function (pmf):** A function defined as  $p_X(x) = P(X = x)$  for values in the range space, and 0 elsewhere.
  - **Examples:** Sum of numbers on two thrown dice, or the number of heads in two coin tosses.
  - **Properties of pmf:** The function must satisfy  $p_X(x) \geq 0$  and  $\sum_{x \in R_X} p_X(x) = 1$ . Any function satisfying these conditions can serve as a pmf.
  - **cdf of a Discrete RV:** Evaluated as the sum of probabilities up to  $x$ , given by  $F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p_X(x_i)$ .
- **4. Continuous Random Variables**

- **Definition:** A random variable  $X$  is continuous if its distribution function  $F_X$  is an absolutely continuous function.
- **Probability Density Function (pdf):** A non-negative function  $f_X$  such that  $F_X(x) = \int_{-\infty}^x f_X(t)dt$  for all  $x \in \mathbb{R}$ .
- **Properties of Continuous RVs:**
  - **Point probability is zero:** For any  $a \in \mathbb{R}$ ,  $P(X = a) = 0$  because  $F_X$  is a continuous function, meaning  $F_X(a) - F_X(a-) = 0$ .
  - **Interval probability equivalency:**  $P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = \int_a^b f_X(x)dx$ .
- **Properties of pdf:** The function must satisfy  $f_X(x) \geq 0$  for all  $x$ , and the total area under the curve is  $\int_{-\infty}^{\infty} f_X(x)dx = 1$ .
- **Relationship to cdf:** By the Fundamental Theorem of Calculus, the pdf is the derivative of the cdf at all points where  $F_X(x)$  is differentiable:  $f_X(x) = \frac{d}{dx}F_X(x)$ .

## Pages 31 - 40: Quantiles, Expectation, and Moments

### • 1. Quantiles, Median, and Mode

- **Quantile:** For a real number  $p$  ( $0 < p < 1$ ), the quantile of  $p$ -th order ( $\xi_p$ ) is defined such that  $P(X \leq \xi_p) \geq p$  and  $P(X < \xi_p) \leq p$ . Also known as the  $100p$ -th percentile.
- **Median:** The specific quantile where  $p = \frac{1}{2}$  ( $\xi_{1/2}$ ). For a continuous random variable, this is the value  $\xi$  where  $\int_{-\infty}^{\xi} f_X(x)dx = \frac{1}{2}$ .
- **Mode:** The value that maximizes the pmf (for discrete) or pdf (for continuous) of  $X$ .
- **Support:** The set of all real numbers  $x$  such that  $p_X(x) > 0$  (discrete) or  $f_X(x) > 0$  (continuous).

### • 2. Mathematical Expectation (Mean)

- **Definition (Discrete):**  $E(X) = \sum_{x \in R_X} x \cdot p_X(x)$ , provided  $\sum |x|p_X(x)$  is finite.
- **Definition (Continuous):**  $E(X) = \int_{-\infty}^{\infty} x \cdot f_X(x)dx$ , provided  $\int_{-\infty}^{\infty} |x|f_X(x)dx$  is finite.
- **Expectation of a Function:** For a function  $g(X)$ ,  $E(g(X)) = \sum g(x)p_X(x)$  or  $\int g(x)f_X(x)dx$ .
- **Properties of Expectation:**
  - $E(c) = c$  (where  $c$  is a constant).
  - $E(X + a) = E(X) + a$ .
  - $E(aX) = aE(X)$ .
  - Linearity:  $E(g_1(X) + g_2(X)) = E(g_1(X)) + E(g_2(X))$ .
  - Order preserving: If  $g_1(x) \leq g_2(x)$ , then  $E(g_1(X)) \leq E(g_2(X))$ .

### • 3. Moments and Variance

- **Moments About Origin:** The  $r$ -th moment about the origin is  $\mu'_r = E(X^r)$ . The first moment  $\mu'_1$  is the mean  $\mu = E(X)$ .
- **Moments About Mean:** The  $r$ -th moment about the mean is  $\mu_r = E[(X - \mu)^r]$ . The first moment about the mean  $\mu_1 = E(X - \mu) = 0$ .
- **Variance:** The second moment about the mean, denoted as  $\text{Var } X$  or  $\sigma_X^2 = E[(X - \mu)^2]$ .
  - **Computation Formula:**  $\text{Var } X = E(X^2) - \mu^2$ .
- **Standard Deviation:** The positive square root of the variance,  $\sigma_X = +\sqrt{\text{Var } X}$ .
- **Properties of Variance:**
  - Unchanged by origin shift:  $\text{Var}(X + a) = \text{Var}(X)$ .
  - Scales quadratically:  $\text{Var}(aX) = a^2 \text{Var}(X)$ . Therefore,  $\sigma_{aX} = |a|\sigma_X$ .
- **4. Moment Generating Function (MGF)**
  - **Definition:** For a real number  $h > 0$ , if  $E(e^{tX})$  exists for all  $t \in (-h, h)$ , the MGF is defined as  $M_X(t) = E(e^{tX})$ .
    - Continuous:  $M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$ .
    - Discrete:  $M_X(t) = \sum_{x \in R_X} e^{tx} p_X(x)$ .
  - **Generating Moments:** By expanding  $e^{tX}$  via Taylor series and taking the expectation, the MGF can be represented as a power series:  $M_X(t) = 1 + t\mu'_1 + \frac{t^2}{2!}\mu'_2 + \cdots + \frac{t^r}{r!}\mu'_r + \cdots$
  - **Derivative Property:** The  $r$ -th derivative of the MGF evaluated at  $t = 0$  yields the  $r$ -th moment about the origin:  $\left[\frac{d^r}{dt^r} M_X(t)\right]_{t=0} = \mu'_r$ .
    - *Example Application:* Given  $f(x) = 3e^{-3x}$  for  $x > 0$ ,  $M_X(t) = \frac{3}{3-t}$ . Differentiating once gives  $E(X) = \mu'_1 = \frac{3}{3^2} = \frac{1}{3}$ . Differentiating twice gives  $E(X^2) = \mu'_2 = \frac{2}{3^2}$ .

## Pages 41 - 50: MGF Identity, Inequalities, Skewness, Kurtosis, and Discrete Distributions

- **1. MGF and Distribution Uniqueness**
  - **Uniqueness Theorem:** An MGF uniquely identifies a probability distribution. If  $M_X(t) = M_Y(t)$  for all  $t$  in an open interval  $-h < t < h$ , then the cumulative distribution functions are equal:  $F_X(z) = F_Y(z)$  for all  $z \in \mathbb{R}$ .
  - **Non-existence:** The MGF does not exist for all random variables. For example, if  $f(x) = \frac{1}{x^2}$  for  $x > 1$ , the integral for  $E(e^{tX})$  diverges to infinity.
  - **Characteristic Function:** Defined as  $\Phi(X) = E(e^{itX})$ . Unlike the MGF, the characteristic function exists for every random variable.
  - **MGF Properties:**
    - Origin Shift:  $M_{X+a}(t) = e^{at} M_X(t)$ .

- Scale Change:  $M_{bX}(t) = M_X(bt)$ .

## • 2. Probability Inequalities

- **Markov's Inequality:** For any non-negative function  $u(X)$  where  $E(u(X))$  exists, and any positive constant  $c$ :

$$P[u(X) \geq c] \leq \frac{E[u(X)]}{c}$$

*Proof concept:* Integrate  $u(x)f(x)$  over the region where  $u(x) \geq c$ , replacing  $u(x)$  with  $c$  to establish the inequality bound.

- **Chebyshev's Inequality:** Let  $X$  have finite variance  $\sigma^2$  and mean  $\mu$ . For any  $k > 0$ :

$$P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}$$

- **Equivalent Form:**  $P[|X - \mu| < k\sigma] \geq 1 - \frac{1}{k^2}$ .
- **Example:** For  $k = 2$ , the probability that  $X$  lies within  $2\sigma$  of the mean ( $\mu - 2\sigma < X < \mu + 2\sigma$ ) is at least  $1 - 1/4 = 3/4$ .

## • 3. Central Moments from Origin Moments

- **Moment about arbitrary point A:**  $\mu'_r = E(X - A)^r$ .
- **Central Moments ( $\mu_r$ ):** Moments about the mean  $\mu$ . Can be expressed in terms of origin moments ( $\mu'_r$  where  $A = 0$ ) using binomial expansion:

$$\mu_r = E[(X - \mu)^r] = \mu'_r - \binom{r}{1}\mu'_{r-1}\mu'_1 + \binom{r}{2}\mu'_{r-2}\mu_1'^2 - \cdots + (-1)^r \mu_1'^r$$

- *Examples:*

- $\mu_2 = \mu'_2 - \mu_1'^2$  (Variance)
- $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$
- $\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4$

## • 4. Measures of Shape (Skewness and Kurtosis)

- **Skewness ( $\gamma_1$ ):** Measures asymmetry.  $\gamma_1 = \frac{\mu_3}{\sigma^3} = \frac{\mu_3}{\mu_2'^{3/2}}$ .
  - $\gamma_1 = 0$ : Symmetric distribution.
  - $\gamma_1 > 0$ : Positively skewed (right tail is longer).
  - $\gamma_1 < 0$ : Negatively skewed (left tail is longer).
- **Kurtosis ( $\gamma_2 / \beta_2$ ):** Measures the "flatness" or "peakedness" of the distribution.  $\beta_2 = \frac{\mu_4}{\mu_2'^2}$ . Excess kurtosis is  $\gamma_2 = \beta_2 - 3$ .
  - $\gamma_2 = 0$  ( $\beta_2 = 3$ ): Mesokurtic (normal distribution).
  - $\gamma_2 > 0$  ( $\beta_2 > 3$ ): Leptokurtic (sharper peak, fatter tails).

- $\gamma_2 < 0$  ( $\beta_2 < 3$ ): Platykurtic (flatter peak, thinner tails).
- **5. Special Discrete Distributions (Introduction)**
  - **Discrete Uniform Distribution:** A random variable has equally likely outcomes over a finite set  $\{x_1, \dots, x_k\}$ .
    - PMF:  $p_X(x) = \frac{1}{k}$ .
    - Mean:  $E(X) = \frac{1}{k} \sum x_i$ .
    - Variance:  $Var(X) = \frac{1}{k} \sum x_i^2 - (\frac{1}{k} \sum x_i)^2$ .
  - **Bernoulli Trials:** Repeated independent trials with only two possible outcomes ("success" and "failure") where the probability of success ( $p$ ) remains constant.
  - **Bernoulli Distribution:** PMF for a single trial ( $x \in \{0, 1\}$ ) is  $p_X(x) = p^x(1 - p)^{1-x}$ .

## Pages 51 - 60: Special Discrete Distributions (Bernoulli, Binomial, Geometric, Negative Binomial)

- **1. Bernoulli Distribution (Continued)**
  - **Properties:**
    - Mean:  $E(X) = p$
    - Second Moment:  $E(X^2) = p$
    - Variance:  $Var(X) = E(X^2) - (E(X))^2 = p - p^2 = p(1 - p) = pq$
    - MGF:  $M_X(t) = E(e^{tX}) = (1 - p) + pe^t = q + pe^t$
- **2. Binomial Distribution**
  - **Definition:** Let there be  $n$  independent Bernoulli trials. If  $X$  represents the number of successes in these  $n$  trials, with  $p = P(\text{success})$  and  $q = 1 - p = P(\text{failure})$ , then  $X$  has a binomial distribution.
  - **PMF:**  $b(x; n, p) = \binom{n}{x} p^x q^{n-x}$  for  $x = 0, 1, 2, \dots, n$ .
    - The terms  $n$  and  $p$  are called the *parameters* of the binomial distribution.
  - **Properties:**
    - Sum of probabilities:  $\sum_{x=0}^n \binom{n}{x} p^x q^{n-x} = (q + p)^n = 1^n = 1$ .
    - Mean:  $E(X) = np$ .
    - Variance:  $Var(X) = npq$ .
    - *Key Trait:* For a binomial random variable,  $Var(X) < E(X)$  (since  $q < 1$ ).
  - **MGF:**  $M_X(t) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x q^{n-x} = (q + pe^t)^n$ .
  - **Example Application:** By comparing an MGF like  $M_X(t) = (\frac{3}{4} + \frac{1}{4}e^t)^{10}$  to the standard formula, one can easily identify  $q = 3/4$ ,  $p = 1/4$ , and  $n = 10$ , allowing quick calculation of mean ( $np = 2.5$ ) and variance ( $npq = 15/8$ ).
- **3. Geometric Distribution**



- **Definition:** In a sequence of Bernoulli trials with success probability  $p$ , let  $X$  represent the number of *failures* that precede the *first success*.
- **PMF:**  $p_X(x) = q^x p$  for  $x = 0, 1, 2, \dots$
- **Properties:**
  - Mean:  $E(X) = \frac{q}{p}$ .
  - Variance:  $Var(X) = \frac{q}{p^2}$ .
- **MGF:**  $M_X(t) = \frac{p}{1 - qe^t}$ .
- **Memoryless Property:** If  $X$  is a geometric random variable, and  $s, t$  are positive integers, then  $P(X \geq s + t \mid X \geq s) = P(X \geq t)$ .
- **4. Negative Binomial Distribution**
  - **Definition:** In a sequence of Bernoulli trials with success probability  $p$ , let  $r$  be a fixed positive integer, and let  $X$  represent the number of *failures* before the  $r$ -th success.
  - **Derivation Logic:** The event  $(X = x)$  occurs if there are exactly  $x$  failures in the first  $(x + r - 1)$  trials, AND the  $(x + r)$ -th trial is a success.
  - **PMF:**  $p_X(x) = \binom{x+r-1}{x} p^r q^x$  for  $x = 0, 1, 2, \dots$ 
    - The terms  $r$  and  $p$  are the parameters of  $X$ .
  - **MGF:**  $M_X(t) = \sum_{x=0}^{\infty} \binom{x+r-1}{x} p^r q^x e^{tx} = p^r (1 - qe^t)^{-r} = \left( \frac{p}{1 - qe^t} \right)^r$ .

## Pages 61 - 70: Negative Binomial Variations and Poisson Distribution

- **1. Negative Binomial Distribution (Derivations & Alternate Form)**
  - **MGF Derivations:** The Mean  $E(X) = \frac{rq}{p}$  and Variance  $Var(X) = \frac{rq}{p^2}$  are formally derived by taking the first and second derivatives of the MGF  $M_X(t) = \left( \frac{p}{1 - qe^t} \right)^r$  and evaluating them at  $t = 0$ .
  - **Alternate Interpretation (Total Trials  $Y$ ):** Instead of counting failures  $X$ , let  $Y$  represent the total number of trials required to achieve the  $r$ -th success, meaning  $Y = X + r$ .
    - **PMF:**  $P(Y = y) = \binom{y-1}{r-1} p^r q^{y-r}$  for  $y = r, r + 1, \dots$
    - **Mean & Variance:**  $E(Y) = \frac{r}{p}$  and  $Var(Y) = \frac{rq}{p^2}$ .
    - **MGF:**  $M_Y(t) = e^{rt} M_X(t) = \left( \frac{pe^t}{1 - qe^t} \right)^r$ .
  - **Relationship to Binomial Distribution:** Let  $X_1$  be a Binomial random variable with parameters  $n, p$  and  $Y$  be a Negative Binomial random variable. The probability of getting  $r$  or more successes in  $n$  trials is exactly equal to the probability that the  $r$ -th success occurs in  $n$  or fewer trials:  $P(X_1 \geq r) = P(Y \leq n)$ .

- **Relationship to Geometric Distribution:** The Geometric distribution is a special case of the Negative Binomial distribution where the target number of successes is  $r = 1$ .
- **2. Poisson Distribution**
  - **Definition:** A discrete random variable  $X$  has a Poisson distribution if its Probability Mass Function (PMF) is given by  $p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$  for  $x = 0, 1, 2, \dots$
  - **Parameter:** The value  $\lambda > 0$  is the single parameter defining the distribution.
  - **Key Statistical Properties:**
    - **Mean:**  $E(X) = \lambda$ .
    - **Second Moment:**  $E(X^2) = \lambda^2 + \lambda$ .
    - **Variance:**  $Var(X) = E(X^2) - (E(X))^2 = \lambda$ .
    - **Third Moment:** The third moment  $\mu_3$  is also equal to  $\lambda$ .
    - *Distinctive Feature:* For a Poisson distribution, the mean and variance are always equal ( $Mean = Variance = \lambda$ ).
  - **Moment Generating Function (MGF):** The MGF is  $M_X(t) = e^{\lambda(e^t - 1)}$ .
    - *Identification Example:* Comparing a given MGF like  $M_X(t) = e^{3(e^t - 1)}$  to the standard formula instantly identifies it as a Poisson distribution with  $\lambda = 3$ , meaning both the mean and variance are 3.
  - **Scaling Property (Application):** Poisson parameters scale linearly with the observation interval. For example, if telephone calls arrive at an average rate of 5 calls per hour, the parameter for a 2-hour duration scales to  $\lambda = 2 \times 5 = 10$ .

## Pages 71 - 80: Poisson Process, Binomial Limits, Uniform and Exponential Distributions

- **1. Poisson Process**
  - The Poisson distribution effectively models many natural phenomena where events occur in the span of time or space.
  - Any process which leads to a Poisson distribution is called a Poisson Process.
  - **Conditions for a Poisson Process** in a time span:
    - a. The probability of exactly one occurrence in a small time interval of length  $h$  is  $\lambda h + o(h)$ . The term  $o(h)$  represents any function of  $h$  such that  $\frac{o(h)}{h} \rightarrow 0$  as  $h \rightarrow 0$ .
    - b. The probability of two or more occurrences in a small time interval  $h$  is  $o(h)$ .
    - c. The number of occurrences in non-overlapping time intervals are independent.
  - If  $Y$  denotes the number of occurrences during a time interval  $t$  under these conditions,  $Y$  follows a Poisson distribution with the parameter  $\lambda t$ .
  - **PMF:**  $P(Y = y) = \frac{e^{-\lambda t} (\lambda t)^y}{y!}$  for  $y = 0, 1, 2, \dots$ .

- **2. Poisson Distribution as a Limiting Case of Binomial Distribution**

- A binomial random variable  $X$  with parameters  $n$  and  $p$  tends to a Poisson distribution under specific conditions.
- **Limiting Conditions:**
  - a.  $n \rightarrow \infty$ .
  - b.  $p \rightarrow 0$ .
  - c.  $\lambda = np$  is a constant.
- **Example Application:** For 1000 Bernoulli trials ( $n = 1000$ ) with a success probability of 0.005 ( $p = 0.005$ ), the Poisson parameter is  $\lambda = np = 5$ . The probability of 4 or fewer successes is calculated as  $P(X \leq 4) = \sum_{x=0}^4 e^{-5} \frac{5^x}{x!}$ .

- **3. Uniform (Rectangular) Distribution**

- A continuous random variable  $X$  is said to have a uniform density if its probability density function (pdf) is constant over an interval. It is also called a rectangular distribution.
- **PDF:**  $f(x) = \frac{1}{b-a}$  for  $a \leq x \leq b$  (where  $a \neq b$ ), and 0 elsewhere.
- **Mean:**  $E(X) = \frac{b+a}{2}$ .
- **Second Moment:**  $E(X^2) = \frac{b^2+ab+a^2}{3}$ .
- **Variance:**  $Var X = \frac{(b-a)^2}{12}$ .
- **MGF:**  $M_X(t) = \frac{e^{tb}-e^{ta}}{t(b-a)}$ .
- **Example Application:** If a point is chosen at random in the interval  $[0, 2]$ , its pdf is  $f(x) = \frac{1}{2}$ . The probability that the point is greater than  $\frac{3}{2}$  is  $\int_{3/2}^2 \frac{1}{2} dx = \frac{1}{4}$ .

- **4. Exponential Distribution (Introduction)**

- A continuous random variable  $X$  has an exponential distribution if its pdf depends on a parameter  $\beta > 0$ .
- **PDF:**  $f(x) = \frac{1}{\beta} e^{-x/\beta}$  for  $x > 0$ , and 0 elsewhere.
- **Mean:** Derived by evaluating  $\int_0^\infty x \frac{1}{\beta} e^{-x/\beta} dx$ , which results in  $E(X) = \beta$ .

## Pages 81 - 90: Exponential and Gamma Distributions

- **1. Exponential Distribution (Continued)**

- **Derivations:**
  - Second Moment:  $E(X^2) = \int_0^\infty x^2 \frac{1}{\beta} e^{-x/\beta} dx = 2\beta^2$ .
  - Variance:  $Var(X) = E(X^2) - (E(X))^2 = 2\beta^2 - \beta^2 = \beta^2$ .
  - MGF:  $M_X(t) = \int_0^\infty e^{tx} \frac{1}{\beta} e^{-x/\beta} dx = \frac{1}{1-\beta t}$  (valid for  $|t| < \frac{1}{\beta}$ ).
- **Identification via MGF:** If a random variable has an MGF like  $M_X(t) = \frac{1}{1-2t}$ , it is identifiable as an exponential distribution with parameter  $\beta = 2$ .
- **Memoryless Property:** For any positive real numbers  $s, t$ :

$$P(X > s + t \mid X > s) = P(X > t)$$

This is a defining characteristic of the exponential distribution, meaning past waiting time does not affect future probability.

- **2. Connection to the Poisson Process**

- **Core Principle:** If occurrences take place according to a Poisson process with an average rate of  $\lambda$  occurrences per unit time, then the waiting time  $T$  between two successive occurrences follows an exponential distribution with parameter  $\beta = \frac{1}{\lambda}$ .
- **Proof Logic:** The event  $(T > t)$  is equivalent to "no occurrence during the interval  $[0, t]$ ". Thus,  $P(T > t) = P(Y = 0) = e^{-\lambda t}$ . The CDF is  $F(t) = 1 - e^{-\lambda t}$ , and differentiating gives the exponential PDF  $f(t) = \lambda e^{-\lambda t}$ .

- **3. Gamma Distribution**

- **Definition:** A continuous random variable  $X$  has a gamma distribution if its PDF is defined by two parameters,  $\alpha > 0$  and  $\beta > 0$ :

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} \quad \text{for } x > 0$$

(and 0 elsewhere).

- **Properties:**

- Valid PDF:  $\int_0^\infty f(x)dx = \frac{1}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} e^{-t} dt = 1$ .
- Mean:  $E(X) = \alpha\beta$ .
- Second Moment:  $E(X^2) = \alpha(\alpha + 1)\beta^2$ .
- Variance:  $Var(X) = E(X^2) - (E(X))^2 = \alpha\beta^2$ .
- MGF:  $M_X(t) = \frac{1}{(1-\beta t)^\alpha}$ .

- **Relationship to Exponential:** The exponential distribution is a special case of the gamma distribution where the shape parameter  $\alpha = 1$ .

## Pages 91 - 100: Gamma-Poisson Relationship, Beta, Chi-Square, and Normal Distributions

- **1. Relationship between Gamma Distribution and Poisson Process**

- **Integral Identity:** Through repeated integration by parts, the integral of a gamma-like function can be expressed as a series:

$$\frac{1}{r!} \int_a^\infty y^r e^{-y} dy = e^{-a} \left[ 1 + \frac{a}{1!} + \frac{a^2}{2!} + \cdots + \frac{a^r}{r!} \right]$$

- **Gamma CDF Evaluation:** Let  $X$  be a gamma random variable with parameters  $\alpha = r$  (a positive integer) and  $\beta = 1/\lambda$ . The CDF is:

$$F(x) = P(X \leq x) = 1 - P(X > x) = 1 - \int_x^\infty \frac{1}{(r-1)!} \lambda (\lambda y)^{r-1} e^{-\lambda y} dy$$

Using the substitution  $u = \lambda y$  and the identity above, this evaluates to:

$$F(x) = 1 - e^{-\lambda x} \sum_{k=0}^{r-1} \frac{(\lambda x)^k}{k!}$$

- **Poisson Process Application:** If  $T$  is the time until the  $r$ -th occurrence in a Poisson process with rate  $\lambda$ , then  $P(T \leq t) = 1 - P(\text{fewer than } r \text{ occurrences in } [0, t])$ .
- **Conclusion:** Since the number of occurrences follows a Poisson distribution with parameter  $\lambda t$ ,  $P(T \leq t) = 1 - \sum_{k=0}^{r-1} e^{-\lambda t} \frac{(\lambda t)^k}{k!}$ . This perfectly matches the Gamma CDF. Thus, **the waiting time until the  $r$ -th occurrence follows a Gamma distribution with  $\alpha = r$  and  $\beta = 1/\lambda$ .**

## • 2. Beta Distribution

- **Definition:** A continuous random variable  $X$  has a beta distribution if its PDF is defined for  $0 \leq x \leq 1$  by:

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

(and 0 elsewhere), where parameters  $\alpha > 0, \beta > 0$ .

- **Beta Function:** The normalizing constant is the beta function:  $B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$ . It relates to the gamma function via  $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ .

- **Properties:**

- **Mean:**  $E(X) = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)} = \frac{\alpha}{\alpha+\beta}$ .
- **Second Moment:**  $E(X^2) = \frac{B(\alpha+2, \beta)}{B(\alpha, \beta)} = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}$ .
- **Variance:**  $Var(X) = E(X^2) - (E(X))^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ .
- **MGF:** Expressed as a series  $M_X(t) = \sum_{r=0}^{\infty} \frac{t^r}{r!} \frac{B(\alpha+r, \beta)}{B(\alpha, \beta)}$ .

## • 3. Chi-Square Distribution

- **Definition:** A special case of the gamma distribution where the parameters are fixed to  $\alpha = \frac{r}{2}$  and  $\beta = 2$ , with  $r$  being a positive integer.
- **Degrees of Freedom:** The integer parameter  $r$  is called the degrees of freedom.
- **PDF:**

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2} \quad \text{for } x > 0$$

◦ **Properties:**

- **Mean:**  $E(X) = \alpha\beta = \left(\frac{r}{2}\right)(2) = r$ .
- **Variance:**  $Var(X) = \alpha\beta^2 = \left(\frac{r}{2}\right)(2^2) = 2r$ .
- **MGF:**  $M_X(t) = \frac{1}{(1-2t)^{r/2}}$ .

- **Example:** An MGF of  $M_X(t) = \frac{1}{(1-2t)^2}$  maps to  $(1-2t)^{-r/2}$ , meaning  $r/2 = 2 \Rightarrow r = 4$ . This is a chi-square distribution with 4 degrees of freedom, mean 4, and variance 8.

• **4. Normal Distribution (Introduction)**

- **Definition:** A continuous random variable  $X$  has a normal distribution if its PDF spans  $-\infty < x < \infty$  and is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$$

- **Parameters:**  $\mu$  (any real number) and  $\sigma > 0$  are the parameters of the normal distribution.

## Pages 101 - 110: Normal Distribution Properties, Standardization, and Applications

• **1. Normal Distribution Proofs and Moments**

- **Valid PDF:** The integral of the normal PDF from  $-\infty$  to  $\infty$  is proven to equal 1 by using the substitution  $z = \frac{x-\mu}{\sigma}$  and leveraging the Gamma function evaluation  $\Gamma(1/2) = \sqrt{\pi}$ .
- **Mean:** The expectation  $E(X)$  is calculated by separating the integral into a constant term  $\mu$  and an odd function of  $z$  ( $ze^{-z^2/2}$ ), which evaluates to 0, proving that  $E(X) = \mu$ .
- **Variance:** The variance is computed via  $Var(X) = E(X - \mu)^2$ . By substituting  $z = \frac{x-\mu}{\sigma}$  and taking  $u = z^2/2$ , it evaluates to  $\sigma^2 \frac{2}{\sqrt{\pi}} \Gamma(3/2) = \sigma^2$ .

• **2. Moment Generating Function (MGF) of Normal Distribution**

- **Derivation:** The MGF  $M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$  is evaluated by substituting  $z = \frac{x-\mu}{\sigma}$ .
- **Completing the Square:** The exponent in the integral is rearranged into a perfect square  $-\frac{1}{2}(z - \sigma t)^2 + \frac{\sigma^2 t^2}{2}$ .
- **Result:** The integral simplifies to yield the MGF:  $M_X(t) = e^{\mu t + \sigma^2 t^2/2}$ .

• **3. Standard Normal Random Variable ( $Z$ )**

- **Standardization:** If  $X$  is a normal random variable with parameters  $\mu$  and  $\sigma^2$ , the transformed variable  $Z = \frac{X-\mu}{\sigma}$  is called a standard normal random variable.
- **Parameters of  $Z$ :** Since  $\mu$  and  $\sigma$  are constants,  $E(Z) = \frac{1}{\sigma}(E(X) - \mu) = 0$  and  $Var(Z) = \frac{1}{\sigma^2} Var(X) = 1$ . Therefore,  $Z \sim N(0, 1)$ .

- **4. Calculating Normal Probabilities**

- **Intervals:** To find the probability  $P(a \leq X \leq b)$ , the bounds are converted to standard  $z$ -scores:  $z_1 = \frac{a-\mu}{\sigma}$  and  $z_2 = \frac{b-\mu}{\sigma}$ .
- **CDF Usage:** The probabilities are determined using the tabulated standard normal CDF  $\Phi(z)$ , where  $P(z_1 \leq Z \leq z_2) = \Phi(z_2) - \Phi(z_1)$ .
- **Symmetry Property:** The normal distribution is symmetric about its mean, meaning  $f(\mu - x) = f(\mu + x)$ . Probabilities for negative  $z$ -values can be found using symmetry, e.g.,  $P(Z < -a) = 1 - \Phi(a)$  for  $a > 0$ .

- **5. Application Examples**

- **Cotton Breaking Strength:** For a normal distribution with  $\mu = 165$  and  $\sigma^2 = 9$  ( $\sigma = 3$ ), finding the defect probability  $P(X < 162)$  translates to  $P(Z < -1)$ , which equals  $1 - 0.8413 = 0.1587$ .
- **Temperature Ranges:** For  $X \sim N(30, 4)$  where  $\sigma = 2$ , finding  $P(28 < X < 33)$  translates to  $P(-1 < Z < 1.5)$ , which is evaluated by splitting the standard normal area into regions:  $\Phi(1) - 0.5 + \Phi(1.5) - 0.5$ .
- **Light Bulb Defects (Finding Limits):** Given  $X \sim N(2500, 75^2)$ , to find a specific limit  $x$  where only 5% are defective ( $P(X < x) = 0.05$ ), one sets up the equation  $P\left(Z < \frac{x-2500}{75}\right) = 0.05$  to solve backwards using the standard normal table.

## Pages 111 - 120: Lognormal Distribution & Two-Dimensional Random Variables

- **1. Lognormal Distribution**

- **Definition:** A random variable  $X$  with a range of positive real numbers has a lognormal distribution if its natural logarithm,  $Y = \log X$ , follows a normal distribution.
- **Parameters:** The distribution relies on the parameters  $\mu$  and  $\sigma^2$  belonging to the corresponding normal random variable  $Y \sim N(\mu, \sigma^2)$ .
- **Probability Calculations:** The probability within an interval is evaluated by transforming to the standard normal CDF ( $\Phi$ ):  $P(a \leq X \leq b) = \Phi\left(\frac{\log b - \mu}{\sigma}\right) - \Phi\left(\frac{\log a - \mu}{\sigma}\right)$ .
- **CDF and PDF:**
  - **CDF:**  $F(x) = P(X \leq x) = \Phi\left(\frac{\log x - \mu}{\sigma}\right)$ .
  - **PDF:** By differentiating the CDF, the density function is  $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma x} e^{-(\log x - \mu)^2 / 2\sigma^2}$  for  $x > 0$  (and 0 elsewhere).
- **Moments:**
  - **Mean:** Derived through integration via substitution ( $z = \frac{\log x - \mu}{\sigma}$ ), yielding  $E(X) = e^{\mu + \sigma^2/2}$ .

- **Second Moment:** Following a similar general integration rule,  $E(X^2) = e^{2\mu+2\sigma^2}$ .
- **Variance:**  $Var(X) = E(X^2) - (E(X))^2 = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$ .
- **2. Two-Dimensional Random Variables (Introduction)**
  - **Definition:** Let  $S$  be a sample space. Two random variables  $X, Y$  assign an ordered pair of real numbers  $(X(s), Y(s))$  to each point  $s \in S$ . This is called a two-dimensional random variable or random vector.
  - **Joint Cumulative Distribution Function (CDF):**
    - Defined as  $F(x, y) = P(X \leq x, Y \leq y)$ .
    - **Properties:**  $F(x, y)$  is right continuous. Its limits are absolute:  $F(-\infty, y) = 0$ ,  $F(x, -\infty) = 0$ , and  $F(\infty, \infty) = 1$ .
    - **Interval Probability:** For a rectangular region where  $a < b$  and  $c < d$ , the probability is  $P(a < X \leq b, c < Y \leq d) = F(b, d) - F(a, d) - F(b, c) + F(a, c)$ .
- **3. Discrete Two-Dimensional Random Variables**
  - **Condition:** The vector  $(X, Y)$  is discrete if its range is a finite or countable set.
  - **Joint Probability Mass Function (PMF):** Defined as  $p(x, y) = P(X = x, Y = y)$ .
  - **Properties of Joint PMF:**
    - a.  $p(x, y) \geq 0$  for all  $(x, y)$  in the range.
    - b. The double summation over the entire range equals 1:  $\sum \sum p(x, y) = 1$ . Any bivariate function satisfying these two conditions can serve as a joint PMF.
  - **Joint CDF Evaluation:**  $F(x, y) = \sum_{s \leq x} \sum_{t \leq y} p(s, t)$ .
  - **Example Application:** In an experiment throwing two dice, if  $X$  is the first die and  $Y$  is the larger of the two numbers, a joint PMF table can be constructed to map the probabilities of every intersecting  $(X, Y)$  outcome.
- **4. Continuous Two-Dimensional Random Variables**
  - **Joint Density Function (PDF):** A bivariate function  $f(x, y)$  serves as the joint probability density function if, for any area  $A$  in  $\mathbb{R}^2$ , the probability is evaluated using a double integral:  $P((X, Y) \in A) = \iint_A f(x, y) dx dy$ .
  - **Rectangular Region:** Specifically,  $P(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f(x, y) dx dy$ .
  - **Properties:** The joint PDF must be non-negative:  $f(x, y) \geq 0$  for all  $x, y \in \mathbb{R}$ .

## Pages 121 - 130: Two-Dimensional Random Variables, Marginals, and Conditional Distributions

- **1. Continuous Two-Dimensional Random Variables (Continued)**
  - **Total Probability:** The total volume under the joint probability density function (PDF) surface must equal 1:  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ .



- **Joint CDF:** If  $F(x, y)$  is the cumulative distribution function, it is evaluated by integrating the joint density up to the given bounds:  $F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) ds dt$ .
- **Extracting PDF from CDF:** The joint density  $f(x, y)$  can be found by taking the mixed partial second derivative of the joint CDF:  $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y) = \frac{\partial^2}{\partial y \partial x} F(x, y)$ .
- **Finding Unknown Constants:** If a joint density contains an unknown constant (e.g.,  $f(x, y) = kx(x - y)$  for  $0 < x < 1, -x < y < x$ ), the constant  $k$  is found by setting the double integral over the specified region equal to 1. In this specific example, evaluating  $\int_0^1 \int_{-x}^x kx(x - y) dy dx = 1$  yields  $k = 2$ .

## • 2. Marginal Distributions (Discrete)

- **Definition:** For a two-dimensional discrete random variable  $(X, Y)$  with a joint probability mass function (PMF)  $p_{X,Y}(x, y)$ , marginal distributions isolate the probabilities of a single variable.
- **Marginal PMF of  $X$ :** Found by summing the joint probabilities over all possible values of  $Y$ :  $p_X(x) = \sum_{y \in R_Y} p_{X,Y}(x, y)$  for all  $x \in R_X$ .
- **Marginal PMF of  $Y$ :** Found by summing the joint probabilities over all possible values of  $X$ :  $p_Y(y) = \sum_{x \in R_X} p_{X,Y}(x, y)$  for all  $y \in R_Y$ .
- **Application:** Marginal probabilities are essentially the row totals or column totals in a joint probability table, and they can be used independently to find expectations, such as  $E(Y)$ .

## • 3. Marginal Densities (Continuous)

- **Definition:** For a two-dimensional continuous random variable  $(X, Y)$  with joint density  $f_{X,Y}(x, y)$ , marginal densities are found by "integrating out" the other variable.
- **Marginal Density of  $X$ :**  $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$ .
- **Marginal Density of  $Y$ :**  $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$ .
- **Probability Calculations:** The probability that  $X$  falls within a specific set  $A$  relies solely on its marginal density:  $P(X \in A) = \int_A f_X(x) dx$ .
- **Example Integration:** Given a joint density  $f_{X,Y}(x, y) = 2$  over the triangular region  $0 < x < 1, 0 < y < x$ :
  - The marginal density of  $X$  is  $f_X(x) = \int_0^x 2 dy = 2x$  for  $0 < x < 1$ .
  - The marginal density of  $Y$  is  $f_Y(y) = \int_y^1 2 dx = 2(1 - y)$  for  $0 < y < 1$ .
  - The expectation  $E(Y)$  using the marginal density is  $\int_0^1 y \cdot 2(1 - y) dy = \frac{1}{3}$ .

## • 4. Conditional Distributions

- **Definition (Discrete):** Let  $(X, Y)$  be a two-dimensional discrete random variable with joint PMF  $p_{X,Y}(x, y)$ . The conditional distribution defines the probability of one variable occurring given a known, fixed value of the other.
- **Conditional PMF of  $X$  given  $Y = y$ :** Defined as  $p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$ , provided the marginal probability  $p_Y(y) > 0$ .

- This function is left undefined for any point  $y$  where  $p_Y(y) = 0$ .
- The logic stems directly from the conditional probability rule:  $P(X = x|Y = y) = \frac{P(X=x \cap Y=y)}{P(Y=y)}$ .
- **Conditional PMF of  $Y$  given  $X = x$ :** Similarly defined as  $p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$ , where  $p_X(x)$  is the marginal PMF of  $X$  and  $p_X(x) > 0$ .

## Pages 131 - 140: Conditional Distributions and Conditional Expectation

### • 1. Conditional Distributions (Discrete)

- **Definition:** For discrete random variables  $X$  and  $Y$  with joint PMF  $p_{X,Y}(x, y)$ , the conditional PMF of  $X$  given  $Y = y$  is  $p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$ , where  $p_Y(y)$  is the marginal PMF of  $Y$  and  $p_Y(y) > 0$ .
- **Valid Probability Distribution:** The conditional PMF  $p_{X|Y}(x|y)$  acts as a valid probability distribution itself because:
  - a.  $p_{X|Y}(x|y) \geq 0$  for all  $x$ .
  - b.  $\sum_{x \in R_X} p_{X|Y}(x|y) = \frac{\sum p_{X,Y}(x,y)}{p_Y(y)} = \frac{p_Y(y)}{p_Y(y)} = 1$ .
- **Calculation Example:** Given a  $3 \times 2$  joint PMF table, to find the conditional distribution of  $X$  given  $Y = 2$ , extract the column where  $Y = 2$ , find its marginal sum ( $p_Y(2) = 3/4$ ), and divide each joint probability in that column by  $3/4$ .

### • 2. Conditional Densities (Continuous)

- **Definition:** For continuous random variables  $X$  and  $Y$  with joint density  $f_{X,Y}(x, y)$ , the conditional density of  $X$  given  $Y = y$  is  $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$ , provided  $f_Y(y) > 0$ .
- **Visualizing the Slice:** Geometrically, evaluating the conditional density  $f_{X|Y}$  at a specific  $y = y_1$  means taking a 2D cross-section or "slice" of the 3D joint density surface along the plane  $y = y_1$ , and then normalizing that slice so its total area equals 1.
- **Calculation Example:** For joint density  $f_{X,Y}(x, y) = 2$  ( $0 < x < 1, 0 < y < x$ ), the marginal density is  $f_Y(y) = 2(1 - y)$ . Therefore, the conditional density  $f_{X|Y}(x|y) = \frac{2}{2(1-y)} = \frac{1}{1-y}$  for  $y < x < 1$ .

### • 3. Conditional Expectation

- **Definition:** Let  $u(X)$  be a function of  $X$ . The conditional expectation of  $u(X)$  given  $Y = y$  is evaluated using the conditional distribution:
  - Discrete:  $E(u(X)|y) = \sum_x u(x)p_{X|Y}(x|y)$ .
  - Continuous:  $E(u(X)|y) = \int_{-\infty}^{\infty} u(x)f_{X|Y}(x|y)dx$ .
- **Conditional Mean:** The specific case where  $u(X) = X$ . It is denoted as  $E(X|y)$ .

- **Conditional Second Moment:** The specific case where  $u(X) = X^2$ . It is denoted as  $E(X^2|y)$ .
- **Example (Discrete):** Using the conditional PMF table derived earlier for  $X$  given  $Y = 1$  (where probabilities are  $2/3, 1/3, 0$  for  $x = -1, 0, 1$ ), the conditional mean is  $E(X|1) = (-1)(2/3) + (0)(1/3) + (1)(0) = -2/3$ .

#### • 4. Conditional Variance

- **Definition:** Evaluates the variance of  $X$  under the restriction that  $Y = y$ .
- **Formula:** Derived similarly to standard variance, but using conditional expectations:

$$Var(X|y) = E(X^2|y) - (E(X|y))^2$$

- **Calculation Example (Continuous):** Given  $f_{X|Y}(x|1/2) = 2$  for  $1/2 < x < 1$ .
  - First conditional moment:  $E(X|1/2) = \int_{1/2}^1 x(2)dx = 3/4$ .
  - Second conditional moment:  $E(X^2|1/2) = \int_{1/2}^1 x^2(2)dx = 7/12$ .
  - Conditional Variance:  $Var(X|1/2) = 7/12 - (3/4)^2 = 7/12 - 9/16 = 1/48$ .

## Pages 141 - 150: Covariance, Correlation, and Independent Random Variables

#### • 1. Covariance

- **Definition:** For two random variables  $X$  and  $Y$ , the covariance measures their joint variability. It is defined as:

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

- **Computation Formula:** By expanding the definition, a simpler formula for calculation is derived:

$$Cov(X, Y) = E(XY) - E(X)E(Y)$$

- **Variance of a Linear Combination:** Let  $Z = aX + bY$  where  $a$  and  $b$  are constants. The variance of  $Z$  incorporates the covariance:

$$Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$$

#### • 2. Correlation Coefficient

- **Definition:** A normalized measure of the linear relationship between two random variables, defined as:

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

- **Bounds:** The correlation coefficient always falls within the range  $-1 \leq \rho_{XY} \leq 1$ .
- **Interpretation:**  $Cov(X, Y)$  and  $\rho_{XY}$  are measures of the linear relationship between  $X$  and  $Y$ .
  - Positive:  $X - \mu_X$  and  $Y - \mu_Y$  tend to have the same sign with high probability.
  - Negative:  $X - \mu_X$  and  $Y - \mu_Y$  tend to have opposite signs with high probability.

### • 3. Independent Random Variables

- **Discrete Condition:**  $X$  and  $Y$  are independent if their joint PMF is the product of their marginal PMFs:

$$p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y) \quad \text{for all } x \in R_X, y \in R_Y$$

- **Continuous Condition:**  $X$  and  $Y$  are independent if their joint PDF is the product of their marginal PDFs:

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y \in \mathbb{R}$$

- **Generalization to  $n$  Variables:** Variables  $X_1, X_2, \dots, X_n$  are independent if  $f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$ .

### • 4. Key Theorems on Independence

- **CDF Equivalence:**  $X$  and  $Y$  are independent if and only if their joint CDF equals the product of their marginal CDFs:  $F_{X,Y}(x, y) = F_X(x)F_Y(y)$  for all  $x, y$ .
- **Interval Probabilities:**  $X$  and  $Y$  are independent if and only if  $P(a < X \leq b, c < Y \leq d) = P(a < X \leq b) \cdot P(c < Y \leq d)$ .
- **Expectation of Functions:** If  $X$  and  $Y$  are independent, and  $u(X)$  and  $v(Y)$  are functions of  $X$  and  $Y$  respectively, then:

$$E[u(X)v(Y)] = E[u(X)] \cdot E[v(Y)]$$

- **Corollaries:** If  $X$  and  $Y$  are independent, then:
  - $E(XY) = E(X)E(Y)$
  - $Cov(X, Y) = 0$

### • 5. Independence vs. Zero Covariance

- **Crucial Distinction:** While independence implies zero covariance ( $Cov(X, Y) = 0$ ), the **converse is NOT true**.
- **Meaning:** Two random variables can have a covariance of 0 but still be dependent. Covariance only measures *linear* relationships, so variables with non-linear dependencies can still yield  $Cov(X, Y) = 0$ .

- **Counter-Example Logic:** In a provided discrete joint PMF table, calculations yield  $E(XY) = 1/4$ ,  $E(X) = 1/3$ , and  $E(Y) = 3/4$ . Thus,  $Cov(X, Y) = 1/4 - (1/3)(3/4) = 0$ . However, checking a specific coordinate,  $p_{X,Y}(-1, 0) = 0$ , but  $p_X(-1) \cdot p_Y(0) = (1/4) \cdot (1/4) = 1/16$ . Since  $0 \neq 1/16$ , the variables are dependent despite having zero covariance.

## Pages 151 - 160: MGF of Sums and Transformations of Random Variables

### • 1. MGF of Sums of Independent Random Variables

- **Theorem:** If  $X$  and  $Y$  are independent random variables, the moment generating function (MGF) of their sum is the product of their individual MGFs:  $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$ .
- **Proof Principle:** Relies on the definition  $M_{X+Y}(t) = E(e^{(X+Y)t})$  and the independence property where the joint density  $f_{XY}(x, y) = f_X(x)f_Y(y)$ , allowing the double integral to separate into two independent integrals.

### • 2. Distributions of Functions of Random Variables (Discrete)

- **One Variable Case:** Let  $X$  be a discrete random variable with PMF  $p_X(x)$ , and  $Y = g(X)$ .
- **PMF of  $Y$ :** The probability mass function of  $Y$  is found by summing the probabilities of all  $x$  values that map to  $y$ :  $p_Y(y) = \sum_{x:g(x)=y} p_X(x)$ .
- **Example:** If  $X$  has values  $-2, -1, 0, 1, 2, 3$ , and  $Y = X^2$ , the support of  $Y$  is  $0, 1, 4, 9$ .  $P(Y = 4) = P(X = -2) + P(X = 2)$ .

### • 3. Continuous Transformations: Distribution Function (CDF) Technique

- **Method:** For a continuous random variable  $X$  and a function  $Y = g(X)$ , the CDF of  $Y$  is first found by evaluating  $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$ .
- **Deriving the PDF:** The density function is then obtained by differentiating the CDF:  $f_Y(y) = \frac{d}{dy} F_Y(y)$ .
- **Example:** Given  $f_X(x) = \frac{1}{2}$  for  $-1 < x < 1$ , the PDF of  $Y = X^2$  is found by evaluating  $F_Y(y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \sqrt{y}$ , which differentiates to  $f_Y(y) = \frac{1}{2\sqrt{y}}$  for  $0 < y < 1$ .

### • 4. Continuous Transformations: One-Variable Formula

- **Theorem:** Let  $y = g(x)$  be a one-to-one differentiable function for all  $x$  in the support of  $X$ .
- **Formula:** The PDF of  $Y$  is given by  $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$ , provided  $\frac{dy}{dx} \neq 0$ .
- **Proof Logic:** Derived by establishing that  $y = g(x)$  must be either strictly increasing or strictly decreasing, determining the CDF bounds, and differentiating. The absolute value

$\left| \frac{dx}{dy} \right|$  accounts for the sign change in the strictly decreasing case.

- **Handling Non-One-to-One Functions:** If  $g(x)$  is not one-to-one over the entire support (e.g.,  $y = x^2$  over  $[-1, 2]$ ), the transformation must be split into multiple separate one-to-one transformations (e.g.,  $x_1 = -\sqrt{y}$  and  $x_2 = \sqrt{y}$ ), and their resulting PDFs are summed together.
- **5. Transformation Technique for Two Random Variables (Jacobian Method)**
  - **Setup:** Let  $X_1, X_2$  be two continuous random variables with joint density  $f_{X_1, X_2}(x_1, x_2)$ . Let  $y_1 = g_1(x_1, x_2)$  and  $y_2 = g_2(x_1, x_2)$  represent a one-to-one transformation.
  - **Formula:** The joint density of  $Y_1, Y_2$  is  $f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}(g_1^{-1}(y_1, y_2), g_2^{-1}(y_1, y_2))|J|$ .
  - **Jacobian ( $J$ ):** The determinant of the matrix of partial derivatives of the inverse transformations with respect to the new variables:  $J = \det \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{pmatrix}$ .
  - **Example:** For  $f(x_1, x_2) = 2$  ( $0 < x_1 < x_2 < 1$ ), transforming to  $y_1 = \frac{x_1}{x_2}$  and  $y_2 = x_2$  yields inverse functions  $x_2 = y_2$  and  $x_1 = y_1 y_2$ . The Jacobian  $J$  evaluates to  $y_2$ .

## Pages 161 - 170: Transformations, Regression Curves, and Least Squares

- **1. Joint Density Transformation (Example Continuation)**
  - **Finding PDF of a Sum:** To find the PDF of  $U = X + Y$  given a joint density (e.g.,  $f(x, y) = 24xy$ ), a dummy variable  $V = Y$  is introduced.
  - **Jacobian Application:** The inverse functions are  $y = v$  and  $x = u - v$ . The Jacobian is  $J = 1$ . The joint density of  $U, V$  is found, and then the marginal density  $f_U(u)$  is calculated by integrating out  $v$  over the newly defined boundaries ( $0 < v < u < 1$ ).
  - **Result:** For this specific example,  $f_U(u) = 4u^3$  for  $0 < u < 1$ .
- **2. Correlation and Regression Curves**
  - **Definition:** The conditional expectation  $E(Y|x)$  is a function of  $x$ . The curve defined by plotting  $E(Y|x)$  as  $x$  varies over its range  $R_X$  is called the **regression curve of  $Y$  on  $X$** .
  - **Reverse Curve:** Similarly,  $E(X|y)$  is the **regression curve of  $X$  on  $Y$** .
  - **Non-Linear Example:** For  $f(x, y) = xe^{-x(1+y)}$ , the conditional density  $f_{Y|X}(y|x)$  evaluates to  $xe^{-xy}$ . The regression curve  $E(Y|x) = \int_0^\infty y(xe^{-xy})dy = \frac{1}{x}$ , which plots as a hyperbola.
- **3. Linear Regression (Theoretical Distribution)**
  - **Theorem:** If the random variables  $X$  and  $Y$  have finite, positive variances, and the regression curve of  $Y$  on  $X$  is strictly linear, then the equation is exactly:

$$E(Y|x) = \mu_y + \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x)$$

- **Theorem (Reverse):** If the regression curve of  $X$  on  $Y$  is linear, it is:

$$E(X|y) = \mu_x + \rho \frac{\sigma_x}{\sigma_y} (y - \mu_y)$$

- **Proof Logic:** Assumes  $E(Y|x) = a + bx$ . By multiplying by the marginal density  $f_X(x)$  and integrating, it proves  $\mu_y = a + b\mu_x$ . Multiplying the original assumption by  $x$ ,  $f_X(x)$  and integrating yields  $Cov(X, Y) + \mu_x\mu_y = a\mu_x + b(\sigma_x^2 + \mu_x^2)$ . Solving this system provides the constants  $a$  and  $b$  in terms of  $\mu$ ,  $\sigma$ , and  $\rho$ .

#### • 4. Linear Regression of Bivariate Data (Method of Least Squares)

- **Objective:** Given  $n$  paired data values  $(x_i, y_i)$ , the goal is to determine the best possible linear relation  $y = a + bx$  to fit the data.
- **Method:** Define the error (residual) as  $e_i = y_i - a - bx_i$ . The best fit line minimizes the sum of the squared errors:

$$S = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$$

- **Minimization:** To find the minimum, set the partial derivatives of  $S$  with respect to  $a$  and  $b$  to zero ( $\frac{\partial S}{\partial a} = 0, \frac{\partial S}{\partial b} = 0$ ).

#### • 5. Least Squares Normal Equations and Solutions

- **Normal Equations:** Taking the derivatives yields two equations:
  - $\sum y_i = na + b \sum x_i \implies \bar{y} = a + b\bar{x}$
  - $\sum x_i y_i = a \sum x_i + b \sum x_i^2$
- **Notation Definitions:**
  - Sum of squares  $X$ :  $S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$
  - Sum of squares  $Y$ :  $S_{yy} = \sum (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2$
  - Sum of products:  $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}$
- **Solutions for Constants:** Solving the normal equations gives:
  - Slope:  $b = \frac{S_{xy}}{S_{xx}}$
  - Intercept:  $a = \bar{y} - b\bar{x}$

#### • 6. Sample Correlation and the Line of Best Fit

- **Sample Variances:** Defined as  $\hat{\sigma}_x^2 = \frac{S_{xx}}{n}$  and  $\hat{\sigma}_y^2 = \frac{S_{yy}}{n}$ .
- **Sample Correlation Coefficient ( $r$ ):**

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{\frac{1}{n} \sum x_i y_i - \bar{x}\bar{y}}{\hat{\sigma}_x \hat{\sigma}_y}$$

- **Line of Best Fit:** The regression line of  $y$  on  $x$  can be elegantly rewritten using  $r$ :

$$y = \bar{y} + r \frac{\hat{\sigma}_y}{\hat{\sigma}_x} (x - \bar{x})$$

- **Reverse Line:** Similarly, the linear regression of  $x$  on  $y$  is:

$$x = \bar{x} + r \frac{\hat{\sigma}_x}{\hat{\sigma}_y} (y - \bar{y})$$

## Pages 171 - 180: Regression, Random Sampling, Law of Large Numbers, and CLT

### • 1. Linear Regression (Application & Example)

- **Practical Calculation:** Demonstrates finding the least squares line  $y = a + bx$  from a dataset.
- **Steps:**
  - a. Calculate sums  $(\sum x_i, \sum y_i, \sum x_i^2, \sum x_i y_i)$  and means  $(\bar{x}, \bar{y})$ .
  - b. Compute variations  $S_{xx}$  and  $S_{xy}$ .
  - c. Solve for slope  $b = \frac{S_{xy}}{S_{xx}}$  and intercept  $a = \bar{y} - b\bar{x}$ .
- **Prediction:** Once established, the regression equation can predict  $y$  for a given  $x$  outside the dataset.

### • 2. Fitting a Quadratic Curve

- **Objective:** Fit data to a non-linear quadratic model:  $y = a + bx + cx^2$ .
- **Method of Least Squares:** Minimize the sum of squared errors  $S = \sum_{i=1}^n (y_i - a - bx_i - cx_i^2)^2$ .
- **Normal Equations:** Setting the partial derivatives of  $S$  with respect to  $a$ ,  $b$ , and  $c$  to zero produces a system of three linear equations:
  - a.  $\sum y_i = na + b \sum x_i + c \sum x_i^2$
  - b.  $\sum x_i y_i = a \sum x_i + b \sum x_i^2 + c \sum x_i^3$
  - c.  $\sum x_i^2 y_i = a \sum x_i^2 + b \sum x_i^3 + c \sum x_i^4$

### • 3. Random Sampling Theory

- **Population:** The set of all values from which a sample is drawn. Characterized by its probability distribution (or density),  $f_X(x)$ .
- **Random Sample:** A set of  $n$  independent and identically distributed (i.i.d.) random variables,  $X_1, X_2, \dots, X_n$ , sharing the population's distribution.
- **Statistic:** Any function evaluated using the random sample (e.g.,  $T(X_1, \dots, X_n)$ ).
- **Sample Mean ( $\bar{X}$ ):** A key statistic defined as  $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ .



- **Moments of the Sample Mean:**
  - Expectation:  $E(\bar{X}) = \mu$  (unbiased estimator of the population mean).
  - Variance:  $Var(\bar{X}) = \frac{\sigma^2}{n}$  (variance decreases as sample size increases).
- **4. The (Weak) Law of Large Numbers (WLLN)**
  - **Theorem:** Let  $X_1, \dots, X_n$  be i.i.d. random variables with finite mean  $\mu$  and variance  $\sigma^2$ . For any positive error margin  $\epsilon$  and confidence threshold  $\delta$  ( $0 < \delta < 1$ ): If  $n \geq \frac{\sigma^2}{\delta\epsilon^2}$ , then  $P(|\bar{X}_n - \mu| < \epsilon) \geq 1 - \delta$ .
  - **Limit Corollary:** Equivalently, as sample size goes to infinity, the probability that the sample mean converges to the true mean is 1:  $\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| < \epsilon) = 1$ .
  - **Proof Logic:** Derived directly by applying Chebyshev's Inequality to the sample mean statistic.
  - **Sample Size Calculation:** Used to determine the minimum  $n$  required to be a certain percentage sure (e.g., 95% or 99%) that  $\bar{X}_n$  is within  $\epsilon$  of  $\mu$ .
- **5. Central Limit Theorem (CLT)**
  - **Core Theorem:** For a sequence of i.i.d. random variables with finite mean  $\mu$  and variance  $\sigma^2$ , the distribution of the standardized sample mean approaches a standard normal distribution  $N(0, 1)$  as  $n \rightarrow \infty$ .
  - **Standardized Forms for Use:**
    - **Using Sample Mean:**  $Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ .
    - **Using Sums:**  $Z = \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} \sim N(0, 1)$ .
  - **Practical Applications:** Allows approximating probabilities for sums or averages of many independent variables, *regardless of their original distribution*.
    - *Example:* Approximating the probability that the sum of 20 independent Poisson random variables exceeds 15 by converting the sum to a  $z$ -score.
    - *Example:* Estimating the probability that the total weight of 100 randomly sampled bolts exceeds a certain threshold.

## Pages 181 - 190: Normal Approximations, Convergence, and Chi-Square Lemmas

- **1. Central Limit Theorem (CLT) Applications**
  - **CLT Example (Exponential):** For a random sample of size  $n = 225$  from an exponential population with mean  $\mu = 4$  and variance  $\sigma^2 = 16$ , the probability that the sample mean exceeds 4.5 is found using the normal approximation:  $P\left(Z > \frac{4.5-4}{4/\sqrt{225}}\right) = 1 - \Phi(1.875)$ .
  - **Normal Approximation of Binomial Distribution:** A binomial random variable  $X$  with parameters  $n$  and  $p$  can be considered the sum of  $n$  independent and identically distributed

Bernoulli random variables.

- **Approximation Formula:** Because  $E(X) = np$  and  $Var(X) = npq$ , the CLT applies such that  $\frac{X - np}{\sqrt{npq}} \sim N(0, 1)$  as  $n \rightarrow \infty$ .

## • 2. Continuity Correction

- **Definition:** When approximating a discrete binomial distribution with a continuous normal distribution, a continuity correction is applied to the bounds.
- **Formula:** For a binomial variable  $X \sim b(n, p)$  and integers  $a \leq b$ , the interval is expanded by 0.5 on both sides:  $P(a \leq X \leq b) = P(a - 0.5 < X < b + 0.5)$ .
- **Standardized Calculation:** This results in  $P\left(\frac{a - 0.5 - np}{\sqrt{npq}} < Z < \frac{b + 0.5 - np}{\sqrt{npq}}\right)$ .
- **Example Application:** Finding the probability of at most 40 loan refusals out of 225 applications (with  $p = 0.20$ ) uses the correction  $P(X \leq 40) = P(X < 40.5)$  before standardizing. The final result is computed using  $\Phi(-0.75)$ .

## • 3. Sampling Distributions

- **Statistic:** A function  $T(X_1, X_2, \dots, X_n)$  of a random sample that does not contain any unknown parameters.
- **Nature of a Statistic:** A statistic is itself a random variable, and its probability distribution is called its sampling distribution.
- **Sample Mean:**  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ .
- **Sample Variance:**  $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ .
- **Sampling Distribution of the Mean:** If the population is normal with known  $\sigma^2$ , then  $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$  exactly. If the population is not normal but  $n$  is large, the same statistic is approximately  $N(0, 1)$ .

## • 4. Types of Convergence

- **Convergence in Probability:** A sequence of random variables  $\{X_n\}$  converges in probability to  $X$  ( $X_n \xrightarrow{P} X$ ) if for every  $\epsilon > 0$ ,  $\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$ .
- **Weak Law of Large Numbers (WLLN):** Stated using this concept, if  $X_1, \dots, X_n$  are i.i.d. with finite mean and variance, the sample mean converges in probability to the population mean:  $\bar{X}_n \xrightarrow{P} \mu$ . This is proven using Chebyshev's inequality as  $n \rightarrow \infty$ .
- **Convergence in Distribution:**  $\{X_n\}$  converges in distribution to  $X$  ( $X_n \xrightarrow{D} X$ ) if  $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$  at all points where the CDF  $F_X(x)$  is continuous.

## • 5. Chi-Square Distribution Lemmas

- **Sum of Independent Chi-Squares:** If  $X_1, \dots, X_n$  are independent Chi-square random variables with  $\nu_1, \dots, \nu_n$  degrees of freedom, their sum  $\sum_{i=1}^n X_i$  is a Chi-square variable with  $\nu_1 + \dots + \nu_n$  degrees of freedom.
- **MGF Proof:** This is proven by multiplying their MGFs:  $\prod \frac{1}{(1-2t)^{\nu_i/2}} = \frac{1}{(1-2t)^{(\sum \nu_i)/2}}$ .

- **Difference of Independent Chi-Squares:** If  $X_1$  (with  $\nu_1$  d.f.) and  $X_2$  are independent, and  $X_1 + X_2$  is Chi-square with  $\nu > \nu_1$  d.f., then  $X_2$  is a Chi-square variable with  $\nu - \nu_1$  degrees of freedom.
- **Standard Normal Squared:** If  $Z \sim N(0, 1)$ , the transformed variable  $Y = Z^2$  has a Chi-square distribution with 1 degree of freedom. This is initiated by defining two one-to-one transformations  $z_1 = -\sqrt{y}$  and  $z_2 = \sqrt{y}$ .

## Pages 191 - 200: Sampling Distributions and Parameter Estimation

### 1. Chi-Square Distribution and Sample Variance

- The variable  $Y = Z^2$  has a Chi-square distribution with 1 degree of freedom when  $Z \sim N(0, 1)$ .
- Let  $S^2$  be the variance of a random sample of size  $n$  drawn from a normal population with variance  $\sigma^2$ .
- The statistic  $\frac{(n-1)S^2}{\sigma^2}$  follows a Chi-square distribution with  $(n - 1)$  degrees of freedom.
- This theorem is proven by decomposing the total sum of squares  $\sum (X_i - \mu)^2$  into  $(n - 1)S^2 + n(\bar{X} - \mu)^2$ .
- Dividing the decomposed equation by  $\sigma^2$  yields the sum of a Chi-square variable with  $n - 1$  degrees of freedom and a Chi-square variable with 1 degree of freedom.

### 2. The t-Distribution (Student's t-Distribution)

- The sampling distribution of the mean when  $\sigma$  is unknown relies on the t-distribution.
- A random variable  $T$  follows a t-distribution with  $\nu$  degrees of freedom if its probability density function is  $f(t) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\pi\nu}\Gamma(\nu/2)} (1 + \frac{t^2}{\nu})^{-(\nu+1)/2}$  for  $-\infty < t < \infty$ .
- The t-distribution is symmetric about its mean,  $E(T) = 0$ .
- For large values of  $\nu$ , the t-distribution approximates the standard normal distribution  $N(0, 1)$ .
- If  $Z \sim N(0, 1)$  and  $Y$  is an independent Chi-square random variable with  $\nu$  degrees of freedom, the ratio  $\frac{Z}{\sqrt{Y/\nu}}$  has a t-distribution with  $\nu$  degrees of freedom.
- For a random sample of size  $n$  from a normal population, the statistic  $\frac{\bar{X} - \mu}{S/\sqrt{n}}$  has a t-distribution with  $n - 1$  degrees of freedom.

### 3. Sampling Distribution of the Difference of Means

- For samples from two normal populations with known variances  $\sigma_1^2$  and  $\sigma_2^2$ , the standardized difference in means  $\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$  follows a standard normal distribution  $N(0, 1)$ .
- The expected value of the difference in sample means is  $E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2$ .
- The variance of the difference in sample means is  $Var(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$ .
- If population variances are unknown but assumed equal ( $\sigma_1 = \sigma_2$ ), the statistic  $\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$  follows a t-distribution with  $n_1 + n_2 - 2$  degrees of freedom.
- The pooled variance is calculated as  $S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$ .
- If variances are unknown, unequal, but the sample sizes are large, the standardized difference of means approximately follows a standard normal distribution.

### 4. Sampling Distribution of Proportions

- For a binomial random variable  $X \sim b(n, p)$ , the standardized form  $\frac{X - np}{\sqrt{npq}}$  approximately follows a standard normal distribution  $N(0, 1)$  for large  $n$ .
- By substituting  $X = n\hat{p}$ , where  $\hat{p}$  is the sample proportion, the statistic  $\frac{\hat{p} - p}{\sqrt{pq/n}}$  approximately follows  $N(0, 1)$  for large  $n$ .

### 5. F-Distribution and Ratio of Variances

- A random variable  $F$  follows an F-distribution with  $(\nu_1, \nu_2)$  degrees of freedom if its probability density function is  $g(f) = \frac{\Gamma((\nu_1 + \nu_2)/2)}{\Gamma(\nu_1/2)\Gamma(\nu_2/2)} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} f^{\nu_1/2 - 1} \left(1 + \frac{\nu_1}{\nu_2} f\right)^{-(\nu_1 + \nu_2)/2}$  for  $f > 0$ .
- If  $X_1$  and  $X_2$  are independent Chi-square random variables with  $\nu_1$  and  $\nu_2$  degrees of freedom, their normalized ratio  $\frac{X_1/\nu_1}{X_2/\nu_2}$  follows an F-distribution with  $(\nu_1, \nu_2)$  degrees of freedom.
- For two random samples from normal populations, the ratio  $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$  follows an F-distribution with  $(n_1 - 1, n_2 - 1)$  degrees of freedom.
- This theorem is proven by representing  $\frac{(n_i - 1)S_i^2}{\sigma_i^2}$  as independent Chi-square variables and applying the foundational F-distribution lemma.

### 6. Parameter Estimation

- Point estimation applies when a random variable's probability density function or probability mass function has a known form but contains unknown parameters.

- This situation is distinguished from cases where the entire probability function is completely unknown.
- An example is a normal distribution where the mean  $\mu$ , the variance  $\sigma^2$ , or both are unknown.
- Another example is a Poisson distribution where the parameter  $\lambda$  is unknown.
- A third example is a Gamma distribution where the parameters  $\alpha$ ,  $\beta$ , or both are unknown.

# Parameter Estimation: Point Estimation

## 1. Fundamentals of Point Estimation

- **Objective:** When a random variable  $X$  has a probability density function  $f(x; \theta)$  or probability mass function  $p(x; \theta)$  where the parameter  $\theta$  is unknown, the goal is to obtain an estimated value of  $\theta$  from a random sample.
- **Estimator:** A statistic  $\hat{\Theta} = T(X_1, X_2, \dots, X_n)$  used to estimate the unknown parameter  $\theta$ .
- **Estimate:** The specific realized value  $\hat{\theta} = T(x_1, x_2, \dots, x_n)$  calculated from a given random sample.
- **Standard Examples:** The sample mean  $\bar{X}$  is an estimator for the population mean  $\mu$ , and the sample variance  $S^2$  is an estimator for the population variance  $\sigma^2$ .

## 2. Unbiased Estimators

- **Definition:** An estimator  $\hat{\Theta}$  is considered unbiased for a population parameter  $\theta$  if its expected value equals the parameter:  $E(\hat{\Theta}) = \theta$ .
- **Sample Mean ( $\bar{X}$ ):** The sample mean is an unbiased estimator of the population mean  $\mu$  because  $E(\bar{X}) = \frac{1}{n}[E(X_1) + \dots + E(X_n)] = \frac{1}{n}[n\mu] = \mu$ .
- **Sample Variance ( $S^2$ ):** The sample variance is an unbiased estimator of the population variance  $\sigma^2$ , proven by demonstrating that  $E(S^2) = \frac{1}{n-1}[\sum E(X_i - \mu)^2 - nE(\bar{X} - \mu)^2] = \frac{1}{n-1}[n\sigma^2 - n\frac{\sigma^2}{n}] = \sigma^2$ .

## 3. Minimum Variance Unbiased Estimator (MVUE)

- **Definition:** An unbiased estimator  $\hat{\Theta}$  is a minimum variance unbiased estimator (or "efficient" estimator) if its variance  $Var(\hat{\Theta})$  is the absolute minimum among the variances of all possible unbiased estimators of  $\theta$ .

- **Cramer-Rao Inequality:** Under general conditions, the variance of an unbiased estimator  $\hat{\Theta}$  is bounded below by:  $Var(\hat{\Theta}) \geq \frac{1}{nE[(\frac{\partial}{\partial\theta} \log f(x;\theta))^2]}$ .
- **Efficiency:** An unbiased estimator is deemed efficient if it satisfies the Cramer-Rao bound with strict equality.
- **Efficiency Ratio:** The efficiency of an unbiased estimator  $\hat{\Theta}$  is the ratio of the Cramer-Rao bound to the actual variance of  $\hat{\Theta}$ :  $Efficiency = \frac{\text{Cramer-Rao bound}}{Var(\hat{\Theta})}$ .
- **Example (Poisson Population):** For a random sample from a Poisson population, the sample mean  $\bar{X}$  is an efficient unbiased estimator of  $\lambda$  because its variance ( $\frac{\lambda}{n}$ ) exactly equals the Cramer-Rao bound derived from  $f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$ .

## 4. Consistent Estimators

- **Definition:** An estimator  $\hat{\Theta}$  is a consistent estimator of a parameter  $\theta$  if it converges in probability to  $\theta$  as the sample size  $n$  approaches infinity ( $\hat{\Theta} \xrightarrow{P} \theta$ ).
- **Mathematical Condition:** For any  $\epsilon > 0$ ,  $\lim_{n \rightarrow \infty} P(|\hat{\Theta} - \theta| < \epsilon) = 1$ .
- **Theorem:** If an estimator  $\hat{\Theta}$  is an unbiased estimator of  $\theta$  and its variance  $Var(\hat{\Theta}) \rightarrow 0$  as  $n \rightarrow \infty$ , then  $\hat{\Theta}$  is a consistent estimator of  $\theta$ .
- **Proof Principle:** This is verified using Chebyshev's inequality, showing that  $P(|\hat{\Theta} - \theta| < \epsilon) \geq 1 - (\sigma_{\hat{\Theta}}^2 / \epsilon^2)$ , which approaches 1 as the variance goes to 0.
- **Example:** For any population with a finite variance  $\sigma^2$ , the sample mean  $\bar{X}$  is a consistent estimator of  $\mu$  because  $E(\bar{X}) = \mu$  and  $Var(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0$  as  $n \rightarrow \infty$ .

## 5. Maximum Likelihood Estimation (MLE) - Introduction

- **Setup:** Involves a population density  $f(x; \theta)$  with an unknown parameter  $\theta$  and a specific realization  $\{x_1, x_2, \dots, x_n\}$  drawn from a random sample.

## Pages 211 - 220: Maximum Likelihood Estimation

- **1. Maximum Likelihood Estimation (MLE) Concept**
  - Let  $x_1, x_2, \dots, x_n$  be a realization of a random sample from a population with a density or probability mass function  $f(x; \theta)$ , where  $\theta$  is an unknown parameter.
  - The likelihood function for  $\theta$  corresponding to the given sample is defined as the product of the individual densities or probabilities:  $L(\theta) = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta)$ .

- An estimate  $\hat{\theta}$  of  $\theta$  is called a maximum likelihood estimate if it maximizes the likelihood function  $L(\theta)$ .
- The corresponding estimator  $\hat{\Theta}$  is formally called the maximum likelihood estimator (MLE) of  $\theta$ .

## • 2. The Log-Likelihood Function

- Because the natural logarithm ( $\log$ ) is a strictly increasing function of  $\theta$ , the value  $\hat{\theta}$  that maximizes the likelihood function  $L(\theta)$  will equivalently maximize the log-likelihood function  $l(\theta) = \log L(\theta)$ .
- The log-likelihood function simplifies calculations by turning products into sums:  $l(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$ .
- If  $l(\theta)$  is a differentiable function with respect to  $\theta$ , the maximum can be found by setting its first derivative to zero:  $\frac{d}{d\theta} l(\theta) = 0$ .

## • 3. Single-Parameter MLE Examples

- **Poisson Distribution:** For a random sample from a Poisson population with parameter  $\lambda$ , the PMF is  $f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$ .
  - The log-likelihood is derived as  $l(\lambda) = -n\lambda + (\sum_{i=1}^n x_i) \log \lambda - \sum_{i=1}^n \log(x_i!)$ .
  - Setting the derivative  $\frac{d}{d\lambda} l(\lambda) = -n + \frac{1}{\lambda} \sum_{i=1}^n x_i = 0$  yields the MLE for  $\lambda$  as the sample mean:  $\hat{\lambda} = \bar{X}$ .
- **Binomial Distribution:** Suppose there are  $x$  successes in  $n$  trials from a binomial distribution with success probability  $\theta$ .
  - The likelihood function is  $L(\theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}$ .
  - Taking the derivative of the log-likelihood gives  $\frac{d}{d\theta} l(\theta) = \frac{x}{\theta} - \frac{n-x}{1-\theta} = 0$ .
  - Solving this equation provides the MLE for  $\theta$  as the sample proportion:  $\hat{\theta} = \frac{x}{n}$ .

## • 4. Multi-Parameter MLE (Normal Distribution)

- If the parameter is a vector  $\theta = (\theta_1, \theta_2, \dots, \theta_k)$ , the MLE is determined by taking partial derivatives with respect to each parameter and setting them to zero:  $\frac{\partial}{\partial \theta_i} l(\theta) = 0$ .
- For a Normal distribution  $N(\mu, \sigma^2)$ , the population density is  $f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma}} e^{-(x-\mu)^2/2\sigma^2}$ .
- The log-likelihood function expands to  $l(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$ .
- **Case 1:  $\sigma^2$  is known:** Solving  $\frac{\partial}{\partial \mu} l(\mu, \sigma^2) = 0$  yields the MLE for the mean as  $\hat{\mu} = \bar{X}$ .
- **Case 2:  $\mu$  is known:** Solving  $\frac{\partial}{\partial \sigma^2} l(\mu, \sigma^2) = 0$  yields the MLE for the variance as  $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$ .
- **Simultaneous Estimation:** Setting both partial derivatives to zero simultaneously gives the joint MLEs:  $\hat{\mu} = \bar{X}$  and  $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2$ .
- *Important Note:* The maximum likelihood estimator derived for  $\sigma^2$  is inherently a biased estimator.

- **5. MLE for Uniform Distribution (Order Statistics)**

- Consider a random sample from a uniform distribution with parameters  $\alpha$  and  $\beta$  ( $\alpha < \beta$ ), where the PDF is  $f(x; \alpha, \beta) = \frac{1}{\beta - \alpha}$  for  $\alpha \leq x \leq \beta$ .
- The likelihood function for the entire sample is  $L(\alpha, \beta) = \frac{1}{(\beta - \alpha)^n}$ .
- To maximize  $L(\alpha, \beta)$ , the denominator  $(\beta - \alpha)$  must be minimized.
- This minimization is subject to the strict constraint that  $\alpha \leq x_i \leq \beta$  for every value in the sample.
- By ordering the sample values such that  $x_{(1)} \leq x_{(2)} \cdots \leq x_{(n)}$ , the constraints simplify to  $\alpha \leq x_{(1)}$  and  $\beta \geq x_{(n)}$ .
- Therefore, the MLEs that minimize  $\beta - \alpha$  are the extreme order statistics:  $\hat{\alpha} = x_{(1)}$  (the minimum sample value) and  $\hat{\beta} = x_{(n)}$  (the maximum sample value).

## Parameter Estimation: Interval Estimation (Pages 221 - 230)

- **1. Concept of Interval Estimation**

- **Contrast with Point Estimation:** Point estimation yields a single value ( $\hat{\theta}$ ) for a population parameter  $\theta$  but provides no information regarding how close this estimate is to the true value.
- **Definition:** Interval estimation estimates a population parameter  $\theta$  within a range or interval:  $\hat{\theta}_1 < \theta < \hat{\theta}_2$ , where  $\hat{\theta}_1$  and  $\hat{\theta}_2$  are values derived from two appropriate estimators  $\hat{\Theta}_1$  and  $\hat{\Theta}_2$ .
- **Confidence Interval:** If  $P(\hat{\Theta}_1 < \theta < \hat{\Theta}_2) = 1 - \alpha$  (for  $0 < \alpha < 1$ ), then the interval  $(\hat{\theta}_1, \hat{\theta}_2)$  is called a  $(1 - \alpha) \times 100\%$  confidence interval for  $\theta$ .
- **Terminology:** The values  $\hat{\theta}_1$  and  $\hat{\theta}_2$  are the confidence limits, and  $(1 - \alpha)$  is the confidence coefficient.
- **Interpretation:** The interval  $\hat{\Theta}_1 < \theta < \hat{\Theta}_2$  is a random interval; the probability equation implies that if many random samples are drawn and intervals computed,  $(1 - \alpha) \times 100\%$  of such intervals will capture the true parameter  $\theta$ .
- **General Derivation:** For a statistic  $T$  with known variance  $\sigma_T^2$  where  $\frac{T - \theta}{\sigma_T} \sim N(0, 1)$ , setting up the probability  $P(-z_{\alpha/2} < \frac{T - \theta}{\sigma_T} < z_{\alpha/2}) = 1 - \alpha$  and rearranging yields the general confidence interval:  $t - z_{\alpha/2}\sigma_T < \theta < t + z_{\alpha/2}\sigma_T$ .

- **2. Confidence Interval for Mean ( $\sigma$  known)**

- **Assumptions:** Let  $\bar{X}$  be the mean of a random sample of size  $n$  from a normal population with mean  $\mu$  and known standard deviation  $\sigma$ .



- **Distribution:** The standardized mean follows a standard normal distribution:  $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ .
- **Formula Derivation:** By solving  $P(-z_{\alpha/2} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < z_{\alpha/2}) = 1 - \alpha$ , the confidence interval for  $\mu$  is derived.
- **Final Formula:** The  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu$  is given by  $\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ .
- **Non-Normal Approximation:** If it is not known that the population is normal, the statistic  $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$  is approximately  $N(0, 1)$  (which improves for larger  $n$ ), making the same formula an approximate confidence interval.
- **Example Calculation:** For a sample of size  $n = 16$  from  $N(\mu, 16)$  yielding  $\bar{x} = 73.8$ , the 95% confidence interval ( $\alpha = 0.05$ ,  $z_{0.025} = 1.96$ ) is calculated as  $73.8 \pm 1.96 \frac{4}{\sqrt{16}}$ , resulting in the interval  $71.84 < \mu < 75.76$ .

### • 3. Confidence Interval for Mean ( $\sigma$ unknown)

- **Assumptions:** Let  $\bar{X}$  be the mean and  $S^2$  be the variance of a random sample of size  $n$  drawn from a normal population with mean  $\mu$ .
- **Distribution:** The statistic  $\frac{\bar{X}-\mu}{S/\sqrt{n}}$  follows a t-distribution with  $(n - 1)$  degrees of freedom.
- **Final Formula:** The  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu$  is given by  $\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$ .
- **Non-Normal Approximation:** If the population is not known to be normal, this same interval serves as an approximate  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu$ , and this approximation improves for large  $n$ .
- **Example Calculation:** For a sample size  $n = 17$  from  $N(\mu, \sigma^2)$  yielding  $\bar{x} = 4.7$  and  $s^2 = 5.76$  ( $s = 2.4$ ), finding the 90% confidence interval involves  $\alpha = 0.10$  and  $t_{0.05, 16} = 1.746$ , substituting these into the formula to find the bounds.

### • 4. Confidence Interval for Difference of Means ( $\sigma_1, \sigma_2$ known)

- **Assumptions:** Let  $\bar{X}_1$  and  $\bar{X}_2$  be the means of two random samples of sizes  $n_1$  and  $n_2$  taken from two normal populations  $N(\mu_1, \sigma_1^2)$  and  $N(\mu_2, \sigma_2^2)$ , where  $\sigma_1^2$  and  $\sigma_2^2$  are known.
- **Distribution:** The standardized difference of the means,  $\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ , follows a standard normal distribution  $N(0, 1)$ .
- **Formula Derivation:** By setting up the probability  $P\left(-z_{\alpha/2} < \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} < z_{\alpha/2}\right) = 1 - \alpha$  and isolating  $(\mu_1 - \mu_2)$ , the confidence limits are derived.

- **Final Formula:** The  $(1 - \alpha) \times 100\%$  confidence interval for the difference  $\mu_1 - \mu_2$  is  $(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ .

# Parameter Estimation: Interval Estimation (Pages 231 - 240)

## 1. Confidence Interval for Difference of Means ( $\sigma_1, \sigma_2$ unknown)

- **Large Sample Approximation:**

- **Condition:** When population variances  $\sigma_1^2$  and  $\sigma_2^2$  are unknown but sample sizes  $n_1$  and  $n_2$  are large.
- **Statistic:** The standardized difference using sample variances  $s_1^2$  and  $s_2^2$  approximately follows a standard normal distribution.
- **Formula:** The approximate  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu_1 - \mu_2$  is:

$$(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

## 2. Difference of Means ( $\sigma_1, \sigma_2$ unknown but equal)

- **Pooled Variance ( $S_p^2$ ):** When it is assumed that the two populations have the same unknown variance ( $\sigma_1^2 = \sigma_2^2 = \sigma^2$ ), the sample variances are combined to form a pooled estimator:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

- $S_p^2$  is proven to be an unbiased estimator of the common population variance  $\sigma^2$  (i.e.,  $E(S_p^2) = \sigma^2$ ).
- **Distribution:** The test statistic combining the sample means and pooled variance follows a  $t$ -distribution with  $(n_1 + n_2 - 2)$  degrees of freedom:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- **Confidence Interval Formula:** The  $(1 - \alpha) \times 100\%$  confidence interval is:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

- **Calculation Example:** Given  $n_1 = 9$ ,  $n_2 = 15$ ,  $\bar{x}_1 = 81.31$ ,  $\bar{x}_2 = 78.61$ , and  $s_1^2 = 60.76$ ,  $s_2^2 = 48.24$ .
  - First, compute  $s_p^2 = \frac{8(60.76) + 14(48.24)}{22} = 52.79 \implies s_p = 7.26$ .
  - Find the critical t-value  $t_{0.025, 22} = 2.074$  and substitute into the bounds equation.

### 3. Confidence Interval for $\mu_1 - \mu_2$ on Paired Data Sets

- **Context:** When observations from two samples are naturally paired (e.g., "before" and "after" measurements on the same subjects), denoted as  $(X_1, Y_1), \dots, (X_n, Y_n)$ .
- **Difference Variable ( $D_i$ ):** Define a new variable  $D_i = X_i - Y_i$ . The sequence  $D_1, D_2, \dots, D_n$  is treated as a random sample of size  $n$  from a normal population  $N(\mu_D, \sigma_D^2)$ .
- **Expected Difference:**  $\mu_D = E(D_i) = E(X_i) - E(Y_i) = \mu_1 - \mu_2$ .
- **Distribution:** The statistic  $\frac{\bar{D} - \mu_D}{s_D/\sqrt{n}}$  has a  $t$ -distribution with  $n - 1$  degrees of freedom.
- **Formula:** The confidence interval for the mean difference  $\mu_D$  (which equals  $\mu_1 - \mu_2$ ) is:

$$\bar{d} - t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}}$$

where  $\bar{d}$  is the sample mean of the differences and  $s_d$  is the sample standard deviation of the differences.

### 4. Confidence Interval for Proportions

- **Context:** Consider a Bernoulli population with a probability of success  $p$ . Let  $X$  be the number of successes in a sample of  $n$  trials, so  $X \sim b(n, p)$ .
- **Sample Proportion ( $\hat{p}$ ):** Defined as  $\hat{p} = \frac{X}{n}$ .
- **Large Sample Approximation:** For large  $n$ , the Central Limit Theorem dictates that the standardized proportion approximately follows a standard normal distribution:

$$\frac{\hat{p} - p}{\sqrt{pq/n}} \sim N(0, 1)$$

- **Estimating Variance:** Since the true population parameter  $p$  (and  $q = 1 - p$ ) in the denominator is unknown, it is approximated using the sample proportions:  $\frac{pq}{n} \approx \frac{\hat{p}\hat{q}}{n}$ .
- **Formula:** The approximate  $(1 - \alpha) \times 100\%$  confidence interval for the population proportion  $p$  is:

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

# Interval Estimation and Hypothesis Testing (Pages 241 - 250)

## 1. Interval Estimation (Continued)

### Confidence Interval for Proportion (Example)

- A 90% confidence interval for a population proportion  $p$  can be determined using a sample proportion  $\hat{p}$ .
- For a sample of size  $n = 100$  and  $x = 30$  successes, the sample proportion is  $\hat{p} = \frac{30}{100} = 0.3$ .
- The confidence interval formula is  $\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$ .

### Confidence Interval for Variance ( $\sigma^2$ )

- Let  $S^2$  be the variance of a random sample of size  $n$  drawn from a normal population with variance  $\sigma^2$ .
- The statistic  $\frac{(n-1)S^2}{\sigma^2}$  follows a chi-square distribution with  $(n - 1)$  degrees of freedom.
- By isolating  $\sigma^2$  in the probability inequality  $P\left(\chi_{1-\alpha/2, n-1}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{\alpha/2, n-1}^2\right) = 1 - \alpha$ , the confidence interval is derived.
- The  $(1 - \alpha) \times 100\%$  confidence interval for  $\sigma^2$  is  $\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}$ , where  $s^2$  is the sample variance value.
- **Example:** For a sample size  $n = 15$  and  $s^2 = 20.42$ , a 98% confidence interval uses  $\alpha = 0.02$  and critical values  $\chi_{0.99, 14}^2 = 29.14$  and  $\chi_{0.01, 14}^2 = 4.66$ . The interval bounds are calculated as  $\frac{14 \times 20.42}{29.14} < \sigma^2 < \frac{14 \times 20.42}{4.66}$ .

## 2. Hypothesis Testing Fundamentals

### Definitions

- A **Statistical Hypothesis** is an assertion about the distribution of one or more random variables.
- A **Parametric Hypothesis** specifically makes an assertion about an unknown parameter  $\theta$  of a population's distribution.

### Types of Hypotheses

- **Simple Hypothesis**: Completely specifies a parameter (e.g.,  $H_0 : \theta = \theta_0$ ).
- **Composite Hypothesis**: Does not completely specify the parameter (e.g.,  $H_1 : \theta \neq \theta_0, \theta > \theta_0$ , or  $\theta < \theta_0$ ).

### Testing Setup

- **Null Hypothesis ( $H_0$ )**: Generally, the hypothesis we want to test about a distribution.
- **Alternative Hypothesis ( $H_1$ )**: The hypothesis paired against the null hypothesis.
- To test  $H_0$ , a random sample is drawn and a **test statistic** is computed based on the sample values.
- The possible values of the parameter are partitioned into two subsets: the **acceptance region** for  $H_0$  and the **rejection region** for  $H_0$ .

## 3. Errors and Power in Hypothesis Testing

### Types of Errors

- **Type I Error**: The error that occurs when rejecting a true null hypothesis.
- **Type II Error**: The error that occurs when accepting a false null hypothesis.

### Probabilities and Regions

- The probability of a Type I error is denoted by  $\alpha$ .
- The probability of a Type II error is denoted by  $\beta$ .
- **Critical Region**: The specific region leading to the rejection of the null hypothesis  $H_0$  is called the critical region of the test.

- **Size:** The size of the critical region is exactly the probability of committing a Type I error ( $\alpha$ ).

## Power of a Test

- Defined as  $1 - \beta$  for  $\theta = \theta_1$ , where  $\theta_1$  is in the region defined by  $H_1$ .
- A test is considered to have more power if the Type II error ( $\beta$ ) is lower.

## 4. Hypothesis Testing Examples

### Evaluating Errors ( $\alpha$ and $\beta$ )

- **Scenario:** A Bernoulli population is tested for  $H_0 : \theta = 0.90$  against  $H_1 : \theta = 0.80$  with a sample size of 50, where  $H_0$  is accepted if successes  $X > 43$ .
- **Type I Error ( $\alpha$ ):** Calculated as  $P(X \leq 43 \mid \theta = 0.90)$ . Using normal approximation and a continuity correction, this becomes  $P(X < 43.5)$ . Standardizing yields 
$$P\left(Z < \frac{43.5 - 50 \times 0.9}{\sqrt{50 \times 0.9 \times 0.1}}\right) = P\left(Z < \frac{-1.5}{\sqrt{4.5}}\right).$$
- **Type II Error ( $\beta$ ):** Calculated as  $P(X > 43 \mid \theta = 0.80)$ , which is  $P(X \geq 44)$  or  $P(X > 43.5)$  with continuity correction. Standardizing yields 
$$P\left(Z > \frac{43.5 - 50 \times 0.80}{\sqrt{50 \times 0.80 \times 0.20}}\right) = P\left(Z > \frac{3.5}{\sqrt{8}}\right).$$

### Finding a Critical Region

- **Scenario:** Testing the mean of a normal population with known variance  $\sigma^2 = 4$ .  $H_0 : \mu = 30$  against  $H_1 : \mu > 30$ .
- **Objective:** Find the threshold  $k$  for the sample mean  $\bar{x} > k$  such that the critical region has size  $\alpha = 0.05$ .
- **Calculation:** This requires setting  $P(\bar{X} > k) = 0.05$  and standardizing to get 
$$P\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > \frac{k - 30}{2/\sqrt{n}}\right) = 0.05.$$

# Hypothesis Testing: Significance, Means, and Errors (Pages 251 - 260)

## 1. Hypothesis Testing: Sample Size and Error Control

- **Determining Sample Size for Specific Errors:**

- When testing a null hypothesis ( $H_0$ ) against an alternative ( $H_1$ ), you can calculate the required minimum sample size  $n$  to guarantee that the probabilities of Type I error ( $\alpha$ ) and Type II error ( $\beta$ ) do not exceed specified thresholds.
- **Example Setup:** Test  $H_0 : \mu = 30$  against  $H_1 : \mu = 32$  with  $\alpha = 0.05$  and target  $\beta \leq 0.01$ .

- **Calculation Logic:**

- Determine the critical boundary  $k$  using  $\alpha$ :  $P(Z > \frac{\sqrt{n}(k-30)}{2}) = 0.05 \implies k = 30 + \frac{2}{\sqrt{n}}(1.645)$ .
- Set up the inequality for  $\beta$ :  $P(\bar{X}_1 < k \mid \mu = 32) \leq 0.01$ .
- Substitute  $k$  and standardize relative to the alternative mean (32):  
$$P\left(Z < \frac{(30 + \frac{2}{\sqrt{n}}(1.645)) - 32}{2/\sqrt{n}}\right) \leq 0.01.$$
- Simplify to  $P(Z < -\sqrt{n} + 1.645) \leq 0.01$ . Using the standard normal table (where  $z_{0.01} = -2.33$ ), solve  $-\sqrt{n} + 1.645 \leq -2.33$  to find the integer sample size  $n$ .

## 2. Tests of Significance

- **Definition:** A statistical test which specifies a simple null hypothesis, a composite alternative hypothesis, and the size of the critical region  $\alpha$ .
- **Level of Significance:** The value  $\alpha$  (the probability of rejecting a true null hypothesis) is called the level of significance for the test.
- **Types of Tails:**
  - **Left Tail Test:**  $H_0 : \theta = \theta_0$  vs.  $H_1 : \theta < \theta_0$ . The critical (rejection) region of area  $\alpha$  is entirely on the left tail of the distribution.
  - **Right Tail Test:**  $H_0 : \theta = \theta_0$  vs.  $H_1 : \theta > \theta_0$ . The critical region of area  $\alpha$  is entirely on the right tail.
  - **Two Tail Test:**  $H_0 : \theta = \theta_0$  vs.  $H_1 : \theta \neq \theta_0$ . The critical region is split evenly ( $\alpha/2$ ) between the left and right tails.

### 3. General Steps for Test of Significance

1. **Formulate** the null hypothesis  $H_0$  and the alternative hypothesis  $H_1$ , and specify the level of significance  $\alpha$ .
2. **Determine** a critical region of size  $\alpha$  using the sampling distribution of an appropriate test statistic.
3. **Compute** the value of the test statistic from the sampled data.
4. **Check** whether the computed test statistic falls within the critical region. Reject  $H_0$  if it does; accept  $H_0$  otherwise.

### 4. Hypothesis Testing of a Single Mean

- **Case 1: Population Variance ( $\sigma^2$ ) Known**
  - **Test Statistic:**  $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ . The computed value is  $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ .
  - **Decision Rules:**
    - Left tail test ( $H_1 : \mu < \mu_0$ ): Reject  $H_0$  if  $z \leq -z_\alpha$ .
    - Right tail test ( $H_1 : \mu > \mu_0$ ): Reject  $H_0$  if  $z \geq z_\alpha$ .
    - Two tail test ( $H_1 : \mu \neq \mu_0$ ): Reject  $H_0$  if  $z \geq z_{\alpha/2}$  or  $z \leq -z_{\alpha/2}$  (i.e.,  $|z| \geq z_{\alpha/2}$ ).
  - **Example:** Testing if the mean of a sample ( $n = 16$ ,  $\bar{x} = 8.4$ ) is significantly less than 10 with  $\sigma = 3.2$  at  $\alpha = 0.05$ .  $z = \frac{8.4 - 10}{3.2/\sqrt{16}} = -2$ . Since  $-2 < -1.645$  (the  $-z_{0.05}$  critical value), reject  $H_0$ .
- **Case 2: Population Variance ( $\sigma^2$ ) Unknown**
  - **Test Statistic:**  $T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$ , which follows a t-distribution with  $(n - 1)$  degrees of freedom. The computed value is  $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ .
  - **Decision Rules:**
    - Left tail test ( $H_1 : \mu < \mu_0$ ): Reject  $H_0$  if  $t \leq -t_{\alpha, n-1}$ .
    - Right tail test ( $H_1 : \mu > \mu_0$ ): Reject  $H_0$  if  $t \geq t_{\alpha, n-1}$ .
    - Two tail test ( $H_1 : \mu \neq \mu_0$ ): Reject  $H_0$  if  $|t| \geq t_{\alpha/2, n-1}$ .
  - **Example:** Sample of  $n = 12$ ,  $\bar{x} = 33.6$ ,  $s = 2.3$ . Test  $\mu = 35$  vs  $\mu < 35$  at  $\alpha = 0.05$ . Compute  $t = \frac{33.6 - 35}{2.3/\sqrt{12}} = -2.1085$ . Critical value  $-t_{0.05, 11} = -1.796$ . Since  $-2.1085 < -1.796$ , reject  $H_0$ .

### 5. Hypothesis Testing of Difference of Two Means

- **Setup:** Test  $H_0 : \mu_1 - \mu_2 = \delta$  against  $H_1$  where  $\mu_1 - \mu_2$  is  $\neq \delta$ ,  $> \delta$ , or  $< \delta$ .
- **Case 1: Variances ( $\sigma_1^2, \sigma_2^2$ ) Known**



- **Test Statistic:** The statistic  $Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$  follows  $N(0, 1)$ .
- **Computed Value:**  $z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$ .
- **Decision Rules:** Uses standard Z-test rules ( $|z| \geq z_{\alpha/2}$  for two-tail,  $z \geq z_{\alpha}$  for right-tail,  $z \leq -z_{\alpha}$  for left-tail).
- **Case 2: Variances ( $\sigma_1^2, \sigma_2^2$ ) Unknown (Large Samples)**
  - **Test Statistic:** If sample sizes  $n_1, n_2$  are large, the sample variances  $s_1^2, s_2^2$  substitute for the population variances.
  - **Approximation:** The statistic  $\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$  is approximately  $N(0, 1)$ .

# Hypothesis Testing: Difference of Means, Proportions, and Variances (Pages 261 - 270)

## 1. Hypothesis Testing: Difference of Two Means (Independent Samples)

- **Case:  $\sigma_1, \sigma_2$  Known (Example Conclusion)**
  - Testing  $H_0 : \mu_1 - \mu_2 = 0.20$  vs.  $H_1 : \mu_1 - \mu_2 \neq 0.20$  using the Z-test statistic.
  - **Calculation:** With  $\alpha = 0.05$ , the computed value is  $z = 1.08$ .
  - **Decision:** Since  $|z| = 1.08 < 1.96 (z_{0.025})$ , it falls in the acceptance region.  $H_0$  is accepted.
- **Case:  $\sigma_1, \sigma_2$  Unknown but Equal**
  - **Test Statistic:** The appropriate statistic uses the pooled sample variance ( $S_p^2$ ) and follows a t-distribution with  $n_1 + n_2 - 2$  degrees of freedom.
  - **Formula:** For  $H_0 : \mu_1 - \mu_2 = \delta$ , the test statistic value is computed as  $t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ .

## 2. Hypothesis Testing: Difference of Means in Paired Data

- **Setup for Paired Data:** Let  $(X_1, Y_1), \dots, (X_n, Y_n)$  be a random sample from a paired data set. Define the difference for each pair as  $D_i = X_i - Y_i$ .

- **Reduction to Single Sample:** The sequence  $D_1, \dots, D_n$  is treated as a single random sample from a normal distribution with mean  $\mu_D = \mu_1 - \mu_2$ .
- **Test Statistic:** The standardized statistic  $\frac{\bar{D} - \mu_D}{S_D / \sqrt{n}}$  has a t-distribution with  $n - 1$  degrees of freedom.
- **Hypothesis Formulation:**
  - $H_0 : \mu_D = \delta$  (Often  $\delta = 0$ , implying  $\mu_1 = \mu_2$ ).
  - $H_1 : \mu_D \neq \delta$  (Two tail),  $\mu_D > \delta$  (Right tail), or  $\mu_D < \delta$  (Left tail).
- **Computed Value:**  $t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{s_d / \sqrt{n}}$ , where  $s_d$  is the sample standard deviation of the differences.

### 3. Hypothesis Testing for a Single Proportion

- **Context:** A random sample of size  $n$  from a Bernoulli population with parameter  $p$ . Let  $X$  be the total number of successes ( $X \sim b(n, p)$ ).
- **Large Sample Approximation:** The standardized variable  $\frac{X - np}{\sqrt{npq}} = \frac{\hat{p} - p}{\sqrt{pq/n}}$  follows an approximate standard normal distribution  $N(0, 1)$  for large  $n$ .
- **Hypothesis Formulation:**  $H_0 : p = p_0$  against  $H_1 : p \neq p_0$ ,  $p > p_0$ , or  $p < p_0$ .
- **Test Statistic Value:**  $z = \frac{x - np_0}{\sqrt{np_0q_0}} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0q_0}{n}}}$ .
- **Application Example:**
  - Testing if the true proportion of people reading a newspaper is  $p = 0.80$  against  $p < 0.80$ , given 75 out of 100 people read it ( $\alpha = 0.01$ ).
  - **Calculation:**  $z = \frac{75 - 80}{\sqrt{16}} = -1.25$ .
  - **Decision:** The critical value for a left-tail test at  $\alpha = 0.01$  is  $-2.33$ . Since  $-1.25 > -2.33$ ,  $H_0$  is accepted.

### 4. Hypothesis Testing of Variance

- **Context:** Let  $S^2$  be the variance of a random sample of size  $n$  from a normal population with variance  $\sigma^2$ .
- **Distribution:** The statistic  $\frac{(n-1)S^2}{\sigma^2}$  follows a Chi-square distribution with  $(n - 1)$  degrees of freedom.
- **Hypothesis Formulation:**  $H_0 : \sigma^2 = \sigma_0^2$  against  $H_1 : \sigma^2 \neq \sigma_0^2$ ,  $\sigma^2 > \sigma_0^2$ , or  $\sigma^2 < \sigma_0^2$ .
- **Test Statistic Value:** Compute  $y = \frac{(n-1)s^2}{\sigma_0^2}$  and reject  $H_0$  if it falls in the corresponding  $\chi^2$  critical region.
- **Two-Tail Example (Bolt Diameters):**
  - Test  $H_0 : \sigma = 0.015$  against  $H_1 : \sigma \neq 0.015$  given  $n = 15$ ,  $s^2 = 0.00011$  at  $\alpha = 0.01$ .

- **Calculation:**  $y = \frac{14(0.00011)}{0.000225} = 6.84$ .
- **Decision:** Critical values are  $\chi^2_{0.005,14} = 4.07$  and  $\chi^2_{0.995,14} = 31.32$ . Since 6.84 is between them,  $H_0$  is accepted.
- **Left-Tail Example (Specific Heat of Iron):**
  - Test  $H_0 : \sigma = 0.0100$  against  $H_1 : \sigma < 0.0100$  given  $n = 9, s = 0.0086$  at  $\alpha = 0.05$ .
  - **Calculation:**  $y = \frac{8(0.00007)}{0.0001} = 5.6$ .
  - **Decision:** Critical value is  $\chi^2_{0.05,8} = 2.73$ . Since  $y = 5.6 > 2.73$ , it does not fall in the left rejection region.  $H_0$  is accepted.

# Master Mind Map: Probability & Statistics (Pages 91 – 270)

## Pages 91 - 100: Advanced Continuous Distributions

- **1. Gamma & Poisson Relationship**
  - The time  $T$  until the  $r$ -th occurrence in a Poisson process (rate  $\lambda$ ) follows a Gamma distribution with  $\alpha = r$  and  $\beta = \frac{1}{\lambda}$ .
- **2. Beta Distribution**
  - **Definition:** PDF is  $f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$  bounded between 0 and 1.
  - **Properties:** Mean  $E(X) = \frac{\alpha}{\alpha+\beta}$ , Variance  $Var X = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ .
- **3. Chi-Square Distribution**
  - **Definition:** A special case of the Gamma distribution where  $\alpha = \frac{r}{2}$  and  $\beta = 2$ .
  - **Properties:** Mean  $E(X) = r$ , Variance  $Var X = 2r$ .
- **4. Normal Distribution (Introduction)**
  - **Definition:** Spans  $-\infty < x < \infty$  with PDF  $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$ .

## Pages 101 - 110: Normal Distribution Properties & Standardization

- **1. Normal Distribution Measures**
  - **Properties:** Mean  $E(X) = \mu$ , Variance  $Var X = \sigma^2$ , MGF  $M_X(t) = e^{\mu t + \sigma^2 t^2/2}$ .
- **2. Standard Normal Random Variable ( $Z$ )**

- **Standardization:** Any normal RV can be standardized using  $Z = \frac{X - \mu}{\sigma}$ , resulting in  $Z \sim N(0, 1)$ .
- **3. Calculating Normal Probabilities**
  - **Symmetry:**  $P(-a \leq Z \leq 0) = P(0 \leq Z \leq a) = \Phi(a) - 0.5$ .
  - **Intervals:**  $P(a \leq X \leq b) = \Phi(z_2) - \Phi(z_1)$ .

## Pages 111 - 120: Lognormal & Two-Dimensional Random Variables

- **1. Lognormal Distribution**
  - **Definition:**  $X$  is lognormal if  $Y = \log X$  has a normal distribution.
  - **Properties:** Mean  $E(X) = e^{\mu + \sigma^2/2}$ , Variance  $Var X = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$ .
- **2. Two-Dimensional Random Variables**
  - **Joint CDF:**  $F(x, y) = P(X \leq x, Y \leq y)$ .
  - **Discrete Joint PMF:**  $p(x, y) = P(X = x, Y = y)$ . Sum over all space equals 1.
  - **Continuous Joint PDF:**  $P((X, Y) \in A) = \iint_A f(x, y) dx dy$ .

## Pages 121 - 130: Joint, Marginal, and Conditional Distributions

- **1. Continuous Joint Distributions**
  - **Relationship:**  $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$ .
- **2. Marginal Distributions**
  - **Discrete:**  $p_X(x) = \sum_y p_{X,Y}(x, y)$ .
  - **Continuous:**  $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$ .
- **3. Conditional Distributions**
  - **Discrete:**  $p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$  (provided  $p_Y(y) > 0$ ).
  - **Continuous:**  $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$ .

## Pages 131 - 140: Conditional Expectations

- **1. Conditional Expectation**

- **Formula:**  $E(u(X)|y) = \int_{-\infty}^{\infty} u(x)f_{X|Y}(x|y)dx$  (or sum for discrete).
- **Conditional Mean:**  $E(X|y)$ .

- **2. Conditional Variance**

- **Formula:**  $Var(X|y) = E(X^2|y) - (E(X|y))^2$ .

## Pages 141 - 150: Covariance, Correlation, and Independence

- **1. Covariance**

- **Definition:**  $Cov(X, Y) = E(XY) - E(X)E(Y)$ .
- **Linear Combinations:**  $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$ .

- **2. Correlation Coefficient**

- **Definition:**  $\rho_{xy} = \frac{Cov(X, Y)}{\sigma_x \sigma_y}$  (Bounded:  $-1 \leq \rho_{xy} \leq 1$ ).

- **3. Independent Random Variables**

- **Condition:**  $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$ .
- **Corollary:** If independent,  $Cov(X, Y) = 0$  (but  $Cov(X, Y) = 0$  does not guarantee independence).

## Pages 151 - 160: Transformations of Random Variables

- **1. MGF of Sums:**  $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$  (if  $X, Y$  are independent).

- **2. Discrete Transformation:**  $p_Y(y) = \sum_{x:g(x)=y} p_X(x)$ .

- **3. Continuous Transformations**

- **CDF Technique:** Find  $F_Y(y) = P(g(X) \leq y)$ , then differentiate.
- **One-Variable Formula:**  $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$ .
- **Jacobian Method (Two-Variable):**  $f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}(x_1, x_2) \cdot |J|$ .

## Pages 161 - 170: Regression and Least Squares

- **1. Regression Curves:** Represents  $E(Y|x)$  or  $E(X|y)$ .
  - **Linear:**  $E(Y|x) = \mu_y + \rho \frac{\sigma_y}{\sigma_x}(x - \mu_x)$ .
- **2. Method of Least Squares**
  - Minimizes sum of squared errors  $S = \sum (y_i - a - bx_i)^2$ .
  - **Slope (b):**  $\frac{S_{xy}}{S_{xx}}$ . **Intercept (a):**  $\bar{y} - b\bar{x}$ .

## Pages 171 - 180: Sampling, LLN, and CLT

- **1. Random Sampling:** A sample of  $n$  independent and identically distributed (i.i.d.) variables.
  - **Sample Mean:**  $E(\bar{X}) = \mu$ ,  $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ .
- **2. Law of Large Numbers (LLN):**  $P(|\bar{X}_n - \mu| < \epsilon) \rightarrow 1$  as  $n \rightarrow \infty$ .
- **3. Central Limit Theorem (CLT):** The standardized sample mean  $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$  converges to  $N(0, 1)$  as  $n \rightarrow \infty$ .

## Pages 181 - 190: Convergence and Sampling Distributions

- **1. Convergence:** Concepts of convergence in probability ( $X_n \xrightarrow{P} X$ ) and distribution ( $X_n \xrightarrow{D} X$ ).
- **2. Chi-Square Additivity:** Sum of independent Chi-square RVs is a Chi-square RV.
- **3. Normal to Chi-Square:** If  $Z \sim N(0, 1)$ , then  $Z^2 \sim \chi^2$  with 1 degree of freedom.

## Pages 191 - 200: Advanced Sampling Distributions ( $t$ and $F$ )

- **1. Sample Variance Distribution:**  $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ .
- **2. Student's  $t$ -Distribution:**  $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ .

- **3. F-Distribution:** Ratio of two independent sample variances  $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1, n_2-1}$ .

## Pages 201 - 210: Parameter Estimation Basics

- **1. Unbiased Estimators:**  $E(\hat{\theta}) = \theta$ . (e.g.,  $\bar{X}$  for  $\mu$ ,  $S^2$  for  $\sigma^2$ ).
- **2. Minimum Variance Unbiased Estimator (MVUE):** Achieves the lowest possible variance among unbiased estimators.
- **3. Cramer-Rao Inequality:** Establishes the lower bound for the variance of an unbiased estimator.
- **4. Consistency:**  $\hat{\theta}_n \xrightarrow{P} \theta$  as  $n \rightarrow \infty$ .

## Pages 211 - 220: Maximum Likelihood Estimation (MLE)

- **1. Likelihood Function:**  $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$ .
- **2. Optimization:** MLEs are found by maximizing  $L(\theta)$ , usually by setting  $\frac{d}{d\theta} \log L(\theta) = 0$ .
- **3. Common MLEs:**
  - Poisson:  $\hat{\lambda} = \bar{X}$ .
  - Binomial:  $\hat{p} = \frac{x}{n}$ .
  - Normal:  $\hat{\mu} = \bar{X}$  and  $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{X})^2$  (Note: MLE for variance is biased).

## Pages 221 - 230: Confidence Intervals

- **1. Single Mean:**
  - $\sigma$  **Known:**  $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ .
  - $\sigma$  **Unknown:**  $\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$ .
- **2. Difference of Means ( $\sigma_1, \sigma_2$  Known):**  $(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ .

## Pages 231 - 240: Advanced Confidence Intervals

- **1. Difference of Means ( $\sigma_1 = \sigma_2$  Unknown):** Utilizes pooled variance  $S_p^2$  and the  $t$ -distribution.
- **2. Paired Data:** Looks at differences  $D_i$ , treating them as a single sample  $t$ -interval framework.
- **3. Proportions:**  $\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$ .

## Pages 241 - 250: Hypothesis Testing Foundations

- **1. Definitions:** Null hypothesis ( $H_0$ ), Alternative hypothesis ( $H_1$ ), Test statistic, Critical region.
- **2. Errors:**
  - **Type I Error ( $\alpha$ ):** Rejecting a true  $H_0$ .
  - **Type II Error ( $\beta$ ):** Accepting a false  $H_0$ .
- **3. Power of a Test:** Defined as  $1 - \beta$ .

## Pages 251 - 260: Tests of Significance

- **1. Tails:** Left-tail ( $<$ ), Right-tail ( $>$ ), Two-tail ( $\neq$ ).
- **2. Single Mean Tests:**
  - **$\sigma$  Known:**  $z$ -test ( $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ ).
  - **$\sigma$  Unknown:**  $t$ -test ( $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ ).

## Pages 261 - 270: Hypothesis Testing for Multiple Parameters

- **1. Difference of Means:**
  - Large samples (unknown variance):  $z$ -test.
  - Small samples (equal unknown variance):  $t$ -test using pooled variance  $S_p$ .
- **2. Paired Data:** Analyzed via a  $t$ -test on the differences  $D_i$ .
- **3. Proportions:** Tested via normal approximation  $z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$ .
- **4. Population Variance:** Tested using a Chi-square statistic  $y = \frac{(n-1)s^2}{\sigma_0^2}$ .